

REVIEW

Electrophoretic deposition of biomaterials

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Electrophoretic deposition (EPD) is attracting increasing attention as an effective technique for the processing of biomaterials, specifically bioactive coatings and biomedical nanostructures. The well-known advantages of EPD for the production of a wide range of microstructures and nanostructures as well as unique and complex material combinations are being exploited, starting from well-dispersed suspensions of biomaterials in particulate form (microsized and nanoscale particles, nanotubes, nanoplatelets). EPD of biological entities such as enzymes, bacteria and cells is also being investigated. The review presents a comprehensive summary and discussion of relevant recent work on EPD describing the specific application of the technique in the processing of several biomaterials, focusing on (i) conventional bioactive (inorganic) coatings, e.g. hydroxyapatite or bioactive glass coatings on orthopaedic implants, and (ii) biomedical nanostructures, including biopolymer–ceramic nanocomposites, carbon nanotube coatings, tissue engineering scaffolds, deposition of proteins and other biological entities for sensors and advanced functional coatings. It is the intention to inform the reader on how EPD has become an important tool in advanced biomaterials processing, as a convenient alternative to conventional methods, and to present the potential of the technique to manipulate and control the deposition of a range of nanomaterials of interest in the biomedical and biotechnology fields.

Keywords: electrophoretic deposition; carbon nanotubes; hydroxyapatite; bioactive glass; scaffolds; tissue engineering

1. INTRODUCTION

Electrophoresis and dielectrophoresis are phenomena of high relevance in biology, biochemistry, materials science, pharmaceutical sciences, biotechnology and chemistry for manipulation of biological materials, e.g. proteins, enzymes, cells, as well as colloids, polymers and solid inorganic particles (Towbin *et al.* 1979; Boehmer 1996; Jensen *et al.* 1998; Lovsky *et al.* 2010). Electrophoretic deposition (EPD) is a special colloidal processing technique that uses the electrophoresis mechanism for the movement of charged particles suspended in a solution under an electric field, to deposit them in an ordered manner on a substrate to develop thin and thick films, coatings and free-standing

bodies (Boehmer 1996; Sarkar & Nicholson 1996; Boccaccini & Zhitomirsky 2002; Besra & Liu 2007). The application of EPD to assemble spherical colloids into highly ordered colloidal crystals is well known (Boehmer 1996; Braun & Wiltzius 1999).

EPD is gaining increasing attention by the materials research community and a wide range of new applications of the technique in the processing of traditional and advanced materials is emerging (Heavens 1990; Sarkar & Nicholson 1996; van der Biest & Vanderperre 1999; Boccaccini & Zhitomirsky 2002; Besra & Liu 2007; Ferrari & Moreno 2010). The interest in EPD is based not only on its high versatility to be used with different materials and combinations of materials but also because EPD is a cost-effective technique usually requiring simple processing equipment and infrastructure. Moreover, EPD has a high potential for scaling up to large product sizes, ranging from micrometres to metres, and it can be adapted to a

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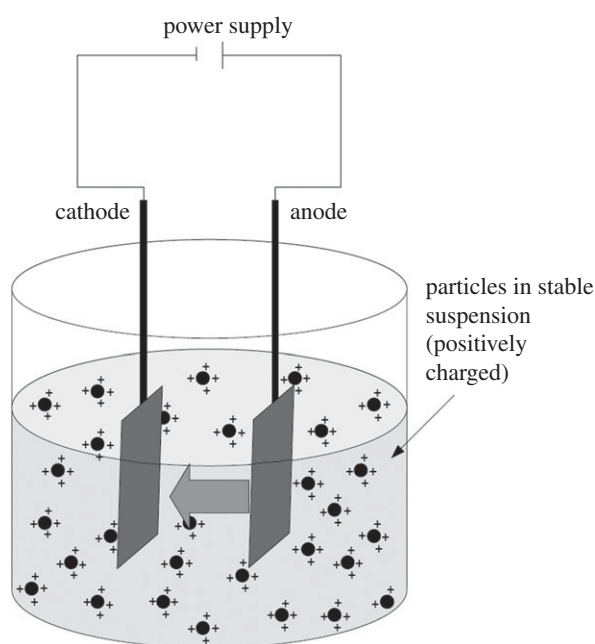


Figure 1. Electrophoretic deposition (EPD) cell showing positively charged particles in suspension migrating towards the negative electrode.

variety of device and component shapes (van der Biest & Vandeperre 1999; Boccaccini & Zhitomirsky 2002; Besra & Liu 2007).

EPD is usually carried out in a two-electrode cell, as schematically shown in figure 1. The motion of charged particles dispersed in a liquid towards the working electrode is achieved by electrophoresis, and the solid deposit formation and growth on the electrode occur primarily via particle coagulation (Heavens 1990; Sarkar & Nicholson 1996). EPD can be applied to a great variety of materials available in the form of fine powders (usually less than approx. 30 μm particle size) or as colloidal suspensions. Metals, polymers, ceramics, glasses and their composites can be deposited by EPD (Gani 1994; van der Biest & Vandeperre 1999).

In the last 15 years, the interest in EPD has widely increased, both in academia and in the industrial sector and numerous new applications for EPD in the development of both bulk materials and coatings are being reported (Besra & Liu 2007), with increasing interest on exploiting the advantages of EPD on nanomaterials (Kanamura & Hamagami 2004; Corni *et al.* 2008) and biological materials (Poortinga *et al.* 2000). Compared with other particle-processing methods, EPD is able to produce uniform deposits (coatings) with high microstructural homogeneity, to provide adequate control of coating thickness and to deposit thin and thick films on substrates of different shapes and on three-dimensional complex and porous structures (van Der Biest & Vandeperre 1999; Kanamura & Hamagami 2004; Besra & Liu 2007; Corni *et al.* 2008). Moreover, the application of EPD to manipulate biological entities and to produce biofilms is being increasingly explored (Poortinga *et al.* 2000; Lovsky *et al.* 2010).

The number of scientific publications that have 'EPD' as a keyword (found using the database Web of Science) has increased notably: from less than 10 papers per year in the 1970s to just under 300 papers published in 2009. This growing interest in EPD in both the academic community and the industrial sector has been accompanied by the establishment of the International Conference series on EPD (with conferences held in 2002, 2005 and 2008 (Boccaccini *et al.* 2009) and the next planned for 2011).

The application of EPD in the biomaterials field started probably with the development of hydroxyapatite (HA) $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$ coatings on Ti substrates in 1986 (Ducheyne *et al.* 1986), gaining further input with the work of Zhitomirsky & Gal-Or (1997), which has also led to investigations of the EPD of HA nanoparticles. The need to develop multifunctional coatings exhibiting strong bonding ability to bone tissue, as well as anti-infectious and anti-allergic properties (Fritsche *et al.* 2009), has given renewed impetus to EPD as the technique of choice for developing nanocomposite coatings. In separate investigations, Roether *et al.* (2002) applied for the first time EPD to coat three-dimensional porous biodegradable polymer (polylactic acid) substrates with Bioglass particles for bone tissue engineering. It was also shown that composite biomaterials, combining polymers and bioactive ceramics, similar to the HAPLEX system developed initially by Bonfield *et al.* (1981) and Wang *et al.* (1998) for orthopaedic applications, can also be fabricated by EPD (Boccaccini *et al.* 2006c). Thus, the applications of EPD in the biomedical sector are being expanded to include a variety of functional, nanostructured and composite coatings, layered and functionally graded biomaterials, thin films, porous biomaterials, tissue scaffolds, drug delivery systems and biosensors, and also for the deposition of biopolymers, bioactive nanoparticles, carbon nanotubes (CNTs) and biological entities (e.g. proteins) in advanced nanostructured biomaterials and devices. In the field of biomaterials, the advantages of EPD mentioned above are highly relevant and determine that EPD, also combined with other electrochemical deposition methods, can be the technique of choice for the development of advanced biomaterial (nano)structures. In this context, EPD can be considered one of the electric field-assisted deposition methods gaining acceptability in the biomaterials field such as electrospraying (Jayasinghe *et al.* 2006; Ahmad *et al.* 2009; De Jonge *et al.* 2009) and electrospinning (Martins *et al.* 2008; Prabhakaran *et al.* 2009).

The intention of this review is to present a comprehensive summary of previous most relevant work on EPD describing the application of the technique in the processing of biomaterials and biomedical structures. Owing to the availability of previous comprehensive review articles covering different aspects of the fundamentals and mechanisms of EPD and its application in the wide materials science field (Heavens 1990; Sarkar & Nicholson 1996; van Der Biest & Vandeperre 1999; Boccaccini & Zhitomirsky 2002; Besra & Liu 2007), the focus of the present article is on the specific developments of EPD in the biomaterials sector only.

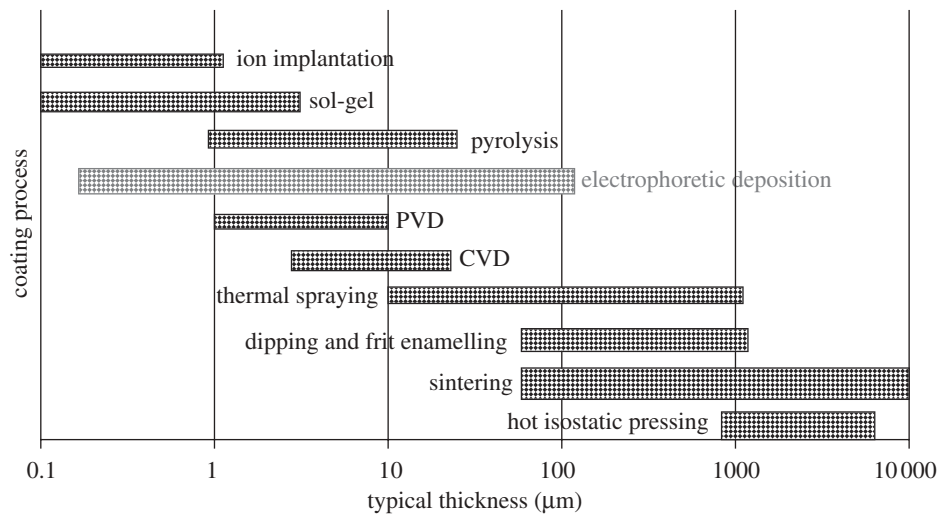


Figure 2. Typical thickness of coatings obtained by different process methods, showing the versatility of EPD in that it can produce a wide range of thicknesses of relevance for orthopaedic applications (Sridhar *et al.* 2002).

2. CONVENTIONAL BIOMEDICAL CERAMIC COATINGS

As a well-established ceramic-coating technique (Jensen *et al.* 1998), the initial and widest application of EPD in the biomedical field has been in the production of inorganic bioactive coatings on metallic substrates for orthopaedic implants. The most important function of a coating in this context is to modify the surface of an implant to improve fixation to the surrounding tissue (Ramaswamy *et al.* 2009). In this regard, bioactive ceramic coatings play a dual role as they prevent the release of potentially harmful metal ions from the metallic substrate (increasing the corrosion resistance of the implant) and they render the surface of the implant bioactive. Bioactive ceramics usually considered for this application are HA, calcium phosphates and silicate bioactive glasses, as discussed next.

2.1. EPD of HA

Synthetic HA is a biologically active calcium phosphate bioceramic commonly used to coat orthopaedic metallic implants or as bone replacement material (Sridhar *et al.* 2002). Bioactive HA promotes bone growth along its surface, and it is an important material for biomedical implants, as its chemical composition is similar to that of the mineral phase of bone (Barralet *et al.* 1998; Suchanek & Yoshimura 1998). Owing to the inferior mechanical properties of HA, however, significant research activity has been associated with the development of HA coatings and composites. HA coatings can promote the attachment of bone tissue and provide a mechanically stable interface between orthopaedic implants and bone. Extensive studies to be discussed in this section have shown that EPD is especially attractive for the deposition of HA on metallic substrates. The interest in EPD to produce HA coatings stems from a variety of reasons such as good stoichiometry control, high purity of the deposited material to a degree not easily achievable by other

processing techniques and the possibility of deposition on substrates of complex shapes. EPD also enables HA impregnation into porous metallic structures, e.g. scaffolds for bone regeneration. In terms of coating thicknesses that can be achieved by EPD, this can vary in a wide range, e.g. between 0.1 μm and more than 100 μm. Figure 2 shows the typical thickness of coatings obtained by different process methods, showing the versatility of EPD in that it can produce a wide range of coating thicknesses of relevance for orthopaedic applications (Sridhar *et al.* 2002).

2.1.1. Substrate materials and EPD parameters. A wide range of metallic substrates have been considered for EPD of HA, such as Ti and Ti6Al4V alloys (Wei *et al.* 1999b; Sridhar *et al.* 2002). EPD has also enabled the HA impregnation of porous meshes (Ducheyne *et al.* 1990; Kim & Ducheyne 1991). Moreover, investigations have also focused on the EPD of HA on stainless steel (Sridhar *et al.* 2003; Guo *et al.* 2007; Javidi *et al.* 2008), aluminium, brass (Esfahanian *et al.* 2005) and on carbon fibres and felts (Zhitomirsky 1998). Numerous investigations have been conducted to date to analyse the electrokinetic behaviour of HA particles in suspension and to predict the EPD yields in different solvents. Zeta-potential measurements (Kowalchuk *et al.* 1993) have shown that the electrokinetic behaviour of HA depends on HA stoichiometry. Positive zeta potentials, in both acidic and alkaline solutions, for example, have been obtained as a result of the presence of Ca^+ , CaOH^+ and $\text{CaH}_2\text{PO}_4^+$ on the particle surface. In addition, as with other ceramic powders in suspension, the electrokinetic properties of HA particles are influenced by the ionic strength of the solution, pH, concentration of particles in suspension as well as particle size and shape. It has been shown that control of the electrokinetic properties may be valuable in the design of HA coatings for increased bone ingrowth. For example, the investigation of the surface properties of calcium-deficient HA powders prepared under different conditions showed that the isoelectric point (IEP)

of HA depended strongly on the synthesis conditions (Barroug *et al.* 1992). IEP values of 4 were found for HA prepared under more acidic conditions, while values ranging from 5.5 to 7.2 were found for HA precipitated from alkaline solutions. These studies highlighted the importance of the surface properties of HA for interaction with proteins and enzymes and for the *in vivo* behaviour of HA implants. Various surface modification techniques have been proposed for the modification of HA particles in order to achieve high suspension stability and high EPD rate (Borum-Nicholas & Wilson 2003). The use of citric acid as a dispersant allowed the deposition of thin films from aqueous and non-aqueous solvents. The films were investigated for the development of hip prosthesis. Polyvinyl alcohol and *N, N*-dimethylformamide have also been added to HA suspensions to improve the adherence and strength of the coatings and to avoid cracking of the deposits upon drying (Meng *et al.* 2006). However, it should be noted that the selection of dispersants, binders and other additives for biomedical applications is limited because some additives are toxic or may have adverse effects in contact with living tissue. There has been previous work on comparing water versus acetone as solvents for EPD of HA on 316L stainless steel (García-Ruiz *et al.* 2006; Vargas *et al.* 2006). The stabilization of HA particles in water can be achieved using 1 wt% of Dispex N40 and 0.001 M KCl. Under this condition the zeta potential of HA in water suspension was -28 mV. In contrast, no additives were required in acetone suspensions. Dense, homogeneous and crack-free HA coatings of sub-micron particles were obtained by applying 5 V for 60 s in water suspensions. Above a DC field of 5 V, the hydrolysis of water seriously impaired coating formation and structural integrity. Homogeneous and crack-free coatings of submicron particles were also obtained in acetone suspensions applying 400–1000 V for 5 s.

Many investigations have been conducted to analyse the influence of the electric field on deposition yield and microstructure of HA films obtained by EPD. The use of high voltages has the advantages of faster deposition rates, shorter deposition times and higher deposit thicknesses (Zhitomirsky & Gal-Or 1997; Mondragon-Cortez & Vargas-Gutiérrez 2004). The investigation of HA powders with a wide particle-size distribution showed preferred deposition of smaller particles at lower voltages (Zhitomirsky & Gal-Or 1997). It was shown that particles with different charge/radius ratios have different electrophoretic mobilities and segregation effects are usually observed during the EPD process. On the other hand, high voltage can promote agglomeration of small particles. Various surface morphologies have been obtained under constant and dynamic voltages (Meng *et al.* 2006, 2008). The increase in the deposition voltage was seen to result in the incorporation of larger particles into the deposits and the development of higher deposit porosity. In another study, Mondragon-Cortez & Vargas-Gutiérrez (2003) investigated the selective deposition of smaller particles from suspensions with a wide particle-size distribution using high voltage (800 V) and short deposition time

(0.5 s). The selective deposition of finer particles during short times at high voltages was attributed to higher electrophoretic mobility of the finer particles (Mondragon-Cortez & Vargas-Gutiérrez 2003).

2.1.2. EPD of HA nanoparticles. The EPD of HA nanoparticles is the subject of intense current experimental efforts (Wei *et al.* 2005). The deposition of nanoparticles offers advantages for the fabrication of ceramic coatings and bodies with dense particle packing ('green' density), good sinterability and homogeneous microstructure. It is now well established that the state of agglomeration of ceramic powders is an important factor in controlling the sintering behaviour. Agglomerate-free structures made from close-packed fine ceramic particles can be densified at lower sintering temperatures, which is relevant for the fabrication of HA coatings on metals and alloys. Fine HA particles can be obtained using a variety of chemical precipitation methods (which will not be discussed in this article). However, as-precipitated HA nanoparticles exhibit a significant tendency to agglomeration. In principle, agglomeration of nanoparticles can be reduced by the use of dispersants. Charged dispersants are also necessary to obtain stable suspensions and to achieve high electrophoretic mobility of the dispersed nanoparticles. However, there is a risk of deposit contamination related to the use of available commercial dispersants. Moreover, the EPD of as-precipitated HA nanoparticles presents difficulties attributed to the water adsorption. It has been found, for example, that adsorbed water can interfere with electrophoretic transport, and this is the reason why EPD of non-calcined HA powders had not been achieved in early studies using non-aqueous solvents (isopropanol; Ducheyne *et al.* 1986). Moreover, the hydrolysis of water can result in high current densities. In this context, thermal analysis of HA powders and EPD experiments have revealed, crucially, the existence of a critical concentration of adsorbed water above which EPD is not possible (Ducheyne *et al.* 1986). Annealing (calcination) of the HA powders at 400°C resulted in dehydration and reduced current during EPD (Ducheyne *et al.* 1986).

The first report on EPD of as-precipitated HA nanoparticles showed that a high deposition rate can be achieved from HA suspensions in isopropanol (Zhitomirsky & Gal-Or 1997). This method was based on a modified wet-route for the fabrication of stoichiometric HA nanoparticles, and the developed approach enabled the preparation of stable suspensions of positively charged HA nanoparticles (Zhitomirsky & Gal-Or 1997). The particle charging and adequate electrostatic stabilization were achieved without additives. As a result, the method led to high purity of the HA deposits. Moreover, the current density was reduced by several orders of magnitude, compared with the current density of greater than 1 A cm^{-2} at an electric field of 90 V cm^{-1} that had been reported earlier (Ducheyne *et al.* 1990). Films prepared at the same electric field but with a much lower current density were of high structural quality, e.g. dense and pinhole-free (Zhitomirsky & Gal-Or 1997).

Table 1. Experimental conditions for EPD of HA nanoparticles.

HA particle size, nm	HA concentration g l ⁻¹	electric field V cm ⁻¹	solvent	reference
55	30	7–130	isopropyl alcohol	Zhitomirsky & Gal-Ohr (1997)
<100	15	330	ethanol	Wei <i>et al.</i> (1999a)
<100	5–10	25–100	acetic anhydride	Wang <i>et al.</i> (2005)
length 50–200, diameter 10–30	100	40	primary aliphatic alcohols	Xiao & Liu (2006)
^a length 20–50, diameter 5–10	20	40	<i>n</i> -butanol-chloroform	Xiao <i>et al.</i> (2008)

^aSi doped HA.

In a series of more recent papers (table 1), the EPD of HA nanoparticles has been investigated under different experimental conditions. Especially interesting are the investigations of the effect of solution ripening of HA nanoparticles on coating morphology (Wei *et al.* 1999a). EPD of unripened HA particles produced highly cracked coatings. It was shown that ambient-ageing ripening for 10 days or boiling for 2 h eliminated cracking in the electrophoretic coatings. Other studies have highlighted the importance of the solvent for the efficient dispersion and deposition of HA nanoparticles with reduced agglomeration and homogeneous microstructure (Wang *et al.* 2005). It was shown, for example, that crack-free and adherent HA coatings can be obtained from stable suspensions of HA nanoparticles (Xiao & Liu 2006; Xiao *et al.* 2008). Also of great interest are the investigations about the effect of substrate surface finishing on the quality of HA coatings deposited by EPD (De Sena *et al.* 2002). Results have shown that surface treatment of titanium samples prior to HA deposition is essential to guarantee high coating adhesion to the metallic substrate. In this context, it should be mentioned that the advantages of using HA nanoparticles may be offset by the fact that nanoparticles may also be more reactive than microparticles and could lead to undesired reactions with the underlying metallic substrates at lower temperature than conventional (micrometric) particles, thus resulting in unstable HA/substrate interfaces.

Research efforts have also been focused on the study of the influence of particle morphology on HA-coating microstructure and properties (Zhang *et al.* 2003b; Kwok *et al.* 2009). Adhesion strength, bioactivity and biocompatibility were studied. In recent experiments, EPD has also been performed using natural HA powder that was extracted from bovine cortical bone (Javidi *et al.* 2008, 2009). The deposition was carried out from HA suspensions in ethanol using polyethylenimine as a binder and dispersing agent. The method resulted in the formation of continuous, uniform and crack-free coatings for applications in biomedical implants. The application of EPD to produce HA films for final use in biosensors has also been investigated recently (Ikoma *et al.* 2009).

2.1.3. Densification of HA EPD coatings and novel EPD approaches to improve coating adhesion. Overall, the analysis of the available literature indicates that

sintering of HA coatings on metallic substrates can result in the degradation of HA and the substrates. The phase transformations in HA upon sintering on Ti and Ti alloy substrates are well documented (Ducheyne *et al.* 1986, 1990; Kim & Ducheyne 1991). Moreover, possible changes in the structure and properties of the metallic substrates are possible. For example, sintering of the HA coating must be performed below the $\alpha + \beta \rightarrow \beta$ transition temperature of Ti–6%Al–4%V substrates (975°C). The use of this alloy requires a high degree of vacuum for the heat treatment of the coatings. On the other hand, sintered HA ceramic has been found to achieve maximum density above 1100°C. Stoichiometric HA is stable in vacuum at temperatures below 1000°C; however, Ti can initiate the decomposition of HA, and vacuum-sintering of HA on Ti alloy substrates can result in undesirable phase transformations. Moreover, it was established that during the sintering of HA coatings, the diffusion of phosphorus into the substrate can result in partial decomposition of HA to form tetra-calcium phosphate with a higher Ca/P ratio. In addition, extensive chemical reactions at the coating/substrate interface usually lead to poor coating adhesion. Several investigations have been carried out to improve the adhesion of HA EPD coatings on metallic alloys. In earlier studies, it was shown that the decomposition rate of HA during sintering can be reduced by the formation of a titanium dioxide layer containing phosphorus (Zhitomirsky & Gal-Or 1997). Such layers can be prepared by anodization of the substrates in phosphoric acid solutions. It was suggested that P-containing oxide layers should provide decreased compositional changes in HA coatings, which result from diffusion of P into the metal substrate. In a similar strategy, a TiO₂ layer can be deposited prior to the deposition of the HA coating (Nie *et al.* 2001; Albayrak *et al.* 2008). It was shown that the use of a TiO₂ interfacial layer resulted in reduced decomposition of HA and improved adhesion of the HA coating. In this regard, the hybrid method based on plasma electrolysis and electrophoresis introduced by Nie *et al.* (2001) may represent the technique of choice to develop HA coatings on titania-covered Ti alloys. A similar approach of the same authors (Nie *et al.* 2000) investigated the deposition of a double-layer HA–TiO₂ coating on Ti alloys by a hybrid treatment involving micro-arc discharge oxidation and EPD.

A dual-coating approach has been developed in order to overcome the difficulties related to the sintering of HA on metallic substrates and to improve the adhesion at the interface (Wei *et al.* 2001). In one experimental study, dual layers of uncalcined HA powder were electrophoretically deposited on Ti, Ti6Al4V and 316L stainless steel substrates and sintered at 875–1000°C. The primary first coating layer exhibited significant cracking. However, the cracking was reduced after the deposition of the second layer. The sintered coatings showed good adhesion to the metallic substrates, and it was found that sintering of the first layer resulted in its chemical degradation (Wei *et al.* 2001). However, this layer acted as a diffusion barrier, which inhibited the metal ions from further migrating from the substrate into the top HA coating.

2.1.4. HA coatings on non-planar structures and patterned HA coatings. In addition to HA coatings on planar substrates, EPD is being investigated as an important tool in the fabrication of HA porous coatings and scaffolds with desired microstructure for various biomedical applications. Porous HA implants with a pore size greater than 50 µm are necessary to meet requirements for osseointegration. Significant interest has been generated in the fabrication of thick coatings with controlled porosity. Highly ordered macroporous bioactive ceramic coatings have been prepared by EPD of polystyrene (PS) spheres (3 µm in diameter) and HA fine particles (200–300 nm in size) from ethanolic suspensions (Hamagami *et al.* 2004, 2006). High-temperature sintering resulted in the burning-out of the PS spheres and the formation of porous films. In related developments, Yousefpour *et al.* (2007) developed porous HA coatings on Ti substrates using polytetrafluoroethylene (PTFE) particles as templates. EPD was carried out at 120 V for 1.5 min using ethanol suspensions of HA (500 nm in size) and PTFE (1–3 µm in size) particles. The composite coatings were heated at 900°C for 60 min to burn out the organic PTFE particles and sinter the HA deposit. Similarly, EPD has been used for the fabrication of hollow uniform HA fibres of controlled diameter (Zhitomirsky 2000). Thick HA layers have also been deposited on graphite rods by repeated depositions (Wang *et al.* 2002). After sintering, the graphite core was burned out and uniform HA tubes were obtained. Porous HA scaffolds with interconnected porosity have been prepared by multiple EPD (Ma *et al.* 2003b) and, more recently, EPD has been used for anisotropic HA formation inside an agarose gel (Watanabe & Akashi 2008).

EPD is also an attractive method to prepare HA-patterned films and coatings (Wang & Hu 2003; Yamaguchi *et al.* 2008), and it can be an alternative technique to other patterning methods developed recently (Li *et al.* 2010b). For example, patterned HA deposits on Si and Ti substrates have been developed for applications in implants and biosensors (Wang & Hu 2003). EPD was used in combination with surface patterning of the cathode, e.g. gold/palladium patterns (hexagons, spherical dots) were created on the cathode,

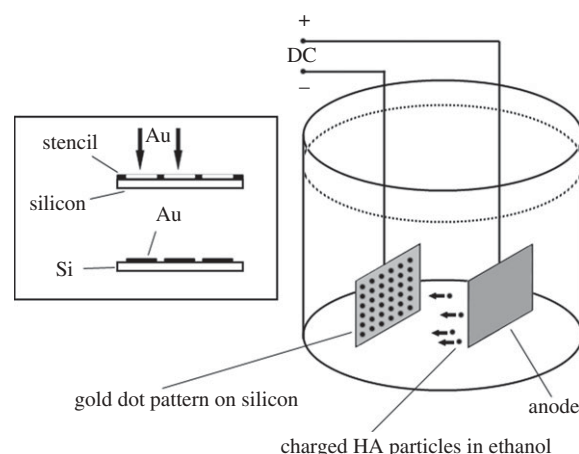


Figure 3. Schematic of the EPD set-up used by Wang & Hu (2003) to develop patterned HA deposits on Si and Ti substrates indicating also the steps followed to prepare the Au-patterned cathode.

and HA colloidal particles in ethanol preferentially deposited on the gold-coated areas forming patterns. The mechanisms for forming HA patterns could involve local variations of the electric field resulting from the presence of the second metal on the cathode. Figure 3 shows a schematic of the EPD set-up used, indicating also the steps followed to prepare the Au-patterned cathode (Wang & Hu 2003). Moreover, EPD has been used for the fabrication of uniform HA coatings on cortical screws for improved bone fixation and reduced risk of interfacial loosening (Yildirim *et al.* 2005) and for achieving HA coatings with enhanced corrosion protection of metallic substrates in simulated body fluid (SBF) (Sridhar *et al.* 2003; Wang *et al.* 2005; Kwok *et al.* 2009). In this regard, EPD enables us to replicate the contour of complex three-dimensional structures (topographies) if the scale of the features is large enough compared with the particles being deposited. Other successful examples of application of EPD to fabricate HA coatings on TiAl6V4 hip prosthesis and on Ti and FeCrAlloy implants have been shown by Mayr *et al.* (2006), Chen *et al.* (2007), Ma *et al.* (2003a) and Guo *et al.* (2007). *In vitro* evaluations using mesenchymal stem cells have shown that cells proliferated more on the HA nanocoating compared with uncoated Ti (Chen *et al.* 2007). Moreover, *in vivo* studies of electrophoretically coated implants for dental applications have shown improved osseointegration (Alzubaydi *et al.* 2009). As expected, in all cases investigated, HA coatings obtained by EPD have led to increased bonding strength at the coated implant–bone interface. It is interesting to point out that doped HA particles, e.g. Si-substituted HA, which have been suggested to exhibit enhanced bioactive and osseointegration effects (Botelho *et al.* 2006), have also been deposited by EPD (Xiao *et al.* 2008).

2.2. EPD of bioactive glasses and glass–ceramics

Bioactive silicate glasses, for example Bioglass of composition (45S5) (in wt%), 45 per cent SiO₂, 24.5 per cent Na₂O, 24.5 per cent CaO and 6 per cent P₂O₅,

were first developed by Hench *et al.* (1971). These materials have been used in numerous biomedical applications such as non-load-bearing bone implants, bioactive coatings for orthopaedics, bone cements and tissue engineering scaffolds (Hench 1998, 2009; Chen *et al.* 2006a) exploiting their excellent bioactivity and biocompatibility. In agreement with the previous discussion on HA, 'bioactivity' in this context refers to the ability of silicate glasses in contact with relevant physiological fluids (body fluid) to form a biologically active hydroxycarbonate apatite (HCA) surface layer that is chemically and structurally equivalent to the mineral phase in bone (Hench 1998). This HCA layer leads to strong interfacial bonding between implants and bone tissue. Bioactive glasses have been applied as coating on biomedical metallic substrates, e.g. Ti alloys and stainless steel, aiming at improving the bioactivity of implants and to protect metallic orthopaedic devices from corrosion when in contact with body fluids (Hench & Andersson 1993).

Recent developments have considered the use of EPD to deposit bioactive glass coatings on metallic substrates as an alternative method to thermal spraying or enamelling. In comparison to HA coatings by EPD, however, much less work has been carried out on EPD of bioactive glass coatings. Krause *et al.* (2006), for example, have claimed to be the first to have investigated the EPD of Bioglass (45S5) powder (of size less than 3 μm) from aqueous suspensions. EPD led to thick (up to 30 μm) bioactive glass deposits on stainless steel. In addition, shape memory nickel–titanium wires were also coated with Bioglass using an optimized EPD procedure (Boccaccini *et al.* 2006a). EPD led to uniform coatings covering the substrates very homogeneously. Best results were achieved with aqueous suspensions containing 20 wt% Bioglass particles, applying voltages of 5 V and deposition time of 5 min (figure 4; Krause *et al.* 2006). For stainless steel substrates, sintering at 950°C led to a complete covering of the plates with a continuous Bioglass layer as a result of viscous flow-assisted densification. Heat treatment of the Bioglass-coated wires at temperatures greater than 800°C led, however, to diffusion of nickel and titanium into the Bioglass coating, which was confirmed by EDX analysis. The results showed that the EPD technique is a very useful method to produce uniform and reproducible Bioglass coatings on metallic planar substrates and wires for biomedical applications. More recent investigations have concentrated on optimizing the EPD process parameters for Bioglass coatings using a design-of-experiments statistical approach based on the Taguchi experimental design method (Pishbin *et al.* 2010). In a separate study, Bibby *et al.* (2003) have investigated novel coatings on Ti6Al4V substrates based on glass–ceramics in the system apatite–mullite (Al_2O_3 – SiO_2 – CaCO_3 – CaF – P_2O_5). Powder of particle size in the range 1–3 μm was used and suspensions for EPD were made using 3 wt% glass powder in water. Hydrochloric acid was added to control the pH. All samples were heat treated in air and vacuum for up to 24 h at 800°C or 1000°C, which led to crystallization of fluoroapatite and mullite of the coating. Thicknesses of approximately 20 μm were obtained.

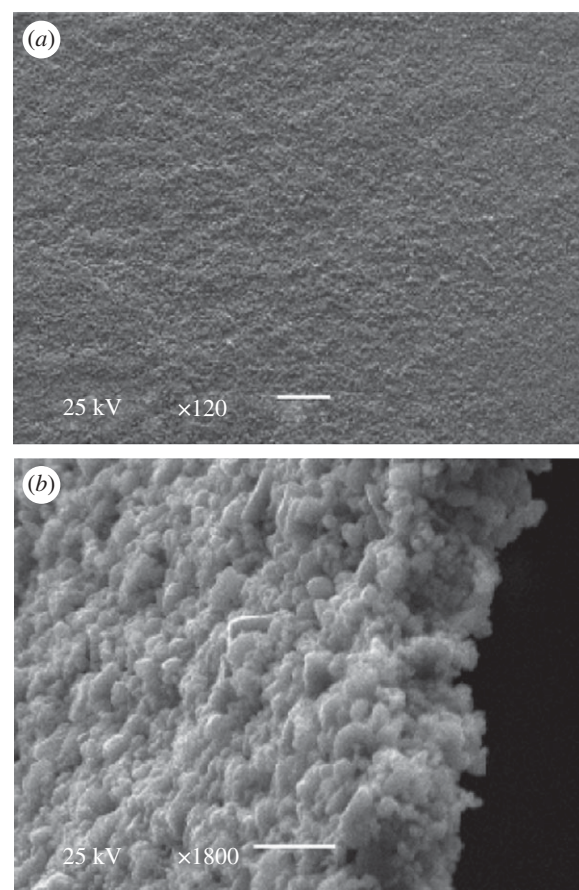


Figure 4. Scanning electron microscope (SEM) images of homogeneous Bioglass coatings obtained on stainless steel substrates by EPD from aqueous suspensions containing 20 wt% Bioglass particles at voltages of 5 V and deposition time of 5 min (Krause *et al.* 2006): (a) top surface and (b) cross section. (Reproduced with permission of Elsevier). Scale bars, (a) 100 μm ; (b) 10 μm .

2.3. EPD of other bioceramic coatings

A few conventional ceramic powders (e.g. other than HA of bioactive glass and not considering nanoparticles, to be discussed in §4.1) have been investigated to produce bioceramic coatings on metallic substrates for biomedical applications. In most cases, the objective has been to apply an oxide coating to act as corrosion protection. For example, Espitia-Cabrera *et al.* (2009) have investigated the EPD of alumina and zirconia films to be used as protective coatings of 316L stainless steel for prostheses and dental implants. The corrosion behaviour in Hanks' solution at room temperature was studied using a potentiodynamic polarization technique. The electrochemical measurements showed a high corrosion resistance of 316L-SS coated by both types of ceramic films but the best behaviour was exhibited by the alumina coating. The morphologies of the corroded films confirmed that the alumina and zirconia films presented low damage with little pitting corrosion compared with the uncoated 316L-SS.

The EPD of CaSiO_3 fine powders (diameter approx. 1.7 μm) on stainless steel was investigated by Hayashi *et al.* (1999). The CaSiO_3 powder was suspended in EtOH – H_2O solvents with different amounts of H_2O (from 0 to 20.2 mass%) to investigate the influence of

H₂O. Similar silicate electrophoretic coatings on stainless steel based on CaMgSi₂O₆ were developed by the same authors (Hayashi *et al.* 2000) from ethanolic suspensions. The corrosion protection capability of CaSiO₃ or CaMgSi₂O₆ coatings has not been investigated, but the coatings could find biomedical applications in the field of bioceramic coatings of metallic implants.

Another interesting bioceramic (incorporating proteins) deposited by EPD on Ti6Al4V has been nacre (mother pearl) (Wang 2004; Zhou *et al.* 2006; Guo & Zhou 2007). Nacre forms the internal layer of many mollusc shells. It has been shown that nacre powder can be a suitable bone-substitute material because of its great similarity to bone mineral, thus applications of this material in the biomedical field have been suggested (Liao *et al.* 1997). The application of EPD to develop coatings at room temperature enables preserving the proteins inside the nacre powder. Suspensions of nacre powder (milled to less than 1 µm) for EPD were prepared by ultrasonication in ethanol (5 wt% suspensions) with addition of acetic acid (Wang 2004). Coating thicknesses of approximately 5 µm were achieved on Ti6Al4V substrates applying an electric field of 100 V cm⁻¹ for 5 min. Patterns of pearl powder on the metallic substrates were obtained using the same method described earlier for HA-patterned coatings (Wang & Hu 2003). In the study of Guo & Zhou (2007), to improve the bonding between the substrate and the coating, the substrate was soaked in a solution of 1.0 mol l⁻¹ H₃PO₄ + 1.5 wt% HF for 20 min to form a TiOx gel on the surface. Three different nacre/ethanol (1.25 g/250 ml) suspensions with and without acid additives were used. EPD was performed at 90 V for 1 min and coatings of sufficient structural quality were fabricated.

2.4. EPD of composite, layered and graded coatings

EPD has also found successful applications in the production of bioactive composite coatings for biomedical implants and devices. This includes both ceramic–ceramic and polymer–ceramic coatings as well as layered and functionally graded coatings.

In the field of ceramic–ceramic coatings for biomedical applications, the main interest has been to develop coatings of HA containing a second phase, e.g. bioactive glass, with the objective of improving the adhesion strength and mechanical properties of the coating without impairing the bioactive behaviour (Maruno *et al.* 1992).

Multi-layered coatings can be conveniently produced via EPD moving the deposition electrode to a second suspension for the deposition of a layer of different compositions when the desired thickness of the first layer has been reached (van Der Biest & Vandeperre 1999). By changing back and forth, a layered material is readily obtained. Moreover, functionally graded materials can also be produced using EPD by gradually changing the composition of the suspension from which EPD is carried out. Early developments in this field have been reviewed by van der Biest & Vandeperre (1999). Of relevance for the biomedical field is the early EPD production of a

range of HA–bioactive glass functionally graded materials (FGMs; Yao *et al.* 2005) and alumina–zirconia-layered systems (Fischer *et al.* 1995; Ferrari *et al.* 1998; Anné *et al.* 2006; Popa *et al.* 2006). More recently, Balamurugan *et al.* (2009) developed functionally graded Bioglass–apatite composite coatings on Ti6Al4V alloy substrates. The coatings were characterized for their relevant properties such as structural integrity and electrochemical behaviour. The electrochemical corrosion parameters such as corrosion potential (E_{corr}) (open circuit potential) and corrosion current density (I_{corr}) evaluated in SBF showed significant shifts towards the noble direction for the graded Bioglass–apatite-coated specimens in comparison with the uncoated Ti6Al4V alloy. The development of bioactive glass–apatite coatings on Ti implants has also been explored by Stojanovic *et al.* (2007). They showed that controlling the deposition voltage and time, the deposition weight and thickness of the produced composite coatings could be controlled with accuracy. Similar HA/bioactive glass composite coatings were developed by Li *et al.* (2001) by electrophoretic codeposition. The graded coatings exhibited high adhesion strength to the substrate after sintering.

Wang *et al.* (2006b) deposited Co–yttria-stabilized zirconia (YSZ)/HA nanocomposite coatings on Ti substrates using the combination of electrocodeposition and EPD. They demonstrated that the Co–YSZ/HA composite coatings exhibited better mechanical properties than nano-HA single coatings. Moreover, the Co–YSZ interlayer reduced the mismatch of the thermal expansion coefficients between HA and Ti and the adhesive strength of the composite coating and the Ti substrate was higher than that of single nano-HA coatings. A related investigation was carried out by Yamashita *et al.* (1997), who investigated the formation of a biomedical ceramic composite of HA and YSZ from ultrasonically mixed suspensions in acetylacetone, 1-propanol and ethanol with a ceramic powder concentration of 2.5–10 g l⁻¹. Moreover, composite bilayer coatings on Ti6Al4V substrates based on biocompatible YSZ in the form of nanoparticles and bioactive Bioglass (45S5) microparticles have also been produced by EPD (Radice *et al.* 2007). In this multi-layered composite approach, the first layer consisted of 5 µm of YSZ, deposited with the intention to avoid possible metal tissue contact. The second layer consisted of 15 µm thick 45S5 Bioglass–YSZ composite, which should react with the surrounding bone tissue to enhance implant fixation. The adsorption of YSZ nanoparticles on bioactive glass microparticles in organic suspension was found to invert the surface charge of the 45S5 Bioglass particles from negative to positive. Thus, cathodic EPD was possible, avoiding uncontrolled anodization (oxidation) of the substrate.

The deposition of composite ceramic coatings on non-metallic substrates for biomedical applications has also been investigated (Ding *et al.* 1995; Yamashita *et al.* 1998). In this case, alumina and zirconia ceramics were coated with apatite composites consisting of powders of HA, alumina and CaO · P₂O₅ with grain sizes between 1 and 3 µm. The powders or their mixtures were dispersed ultrasonically in ethanol, acetylacetone

or a mixture of both for an hour to a concentration of 2.5 g l^{-1} . The ceramic substrates, sputtered with carbon, were placed at the anode and EPD was carried out at 30, 60 and 90 V for 30 s to 6 min.

Another example of application of EPD to deposit composite coatings was shown recently by Guo *et al.* (2008). They developed HA/ Fe_3O_4 composite coatings with hierarchically porous structures by first depositing $\text{CaCO}_3/\text{Fe}_3\text{O}_4$ particles on Ti6Al4V substrates followed by treatment in phosphate buffer saline (PBS) solution at 37°C . The effects of Fe_3O_4 on the conversion rate of calcium carbonate to HCA and the *in vitro* bioactivity of the coatings were investigated. As the $\text{CaCO}_3/\text{Fe}_3\text{O}_4$ coatings are converted to HA/ Fe_3O_4 coatings, macropores with a pore size of approximately $4 \mu\text{m}$ in the coating structure and mesopores with a pore size of approximately 3.9 nm within the HCA plates were formed. Recently, the formation of HA within the porous TiO_2 layer by micro-arc oxidation coupled with EPD has been shown by Kim *et al.* (2009). Addition of ethanol to the electrolyte solution containing fine HA particles was essential to reduce the level of gaseous emission on the anode, which obstructs the attachment of HA particles. *In vitro* cellular assays showed that the incorporation of HA significantly improved the osteoblastic activity on the coating layer, as expected.

Composite coatings involving bioceramics other than HA or bioactive glasses have been developed by Rodriguez *et al.* (2008). They investigated the EPD of wollastonite and porcelain–wollastonite coatings on stainless steel using acetone as the dispersive medium. A direct electric current of 800 V for 3 min was used for obtaining the single wollastonite coating. The two-layer coating was obtained by depositing dental porcelain at 400 V for 30 s followed by the deposition of wollastonite at 400 V for 3 min. After forming the two layers, this complex coating was heat treated at 800°C for 5 min. Strong bonds of both the interface wollastonite–porcelain and that of porcelain–metallic substrate were observed.

There is increasing interest in the development of polymer–bioceramic coatings for orthopaedic applications. The presence of the polymer enables adherence to the metallic substrate at low temperatures, avoiding the problems associated with high-temperature sintering of monolithic ceramic coatings mentioned above. EPD represents a convenient technique to deposit such composite coatings. For example, polyetheretherketone (PEEK)/bioactive glass composite coatings on shape memory alloy (NiTi, Nitinol) wires have been successfully achieved by codeposition from suspensions of PEEK powder (1–6 wt%) and Bioglass particles (0.5–2 wt%) in ethanol using a deposition time of 5 min and applied voltage of 20 V (Boccaccini *et al.* 2006c). EPD led to high quality PEEK/Bioglass coatings with a homogeneous microstructure along the wire length and a uniform thickness of up to $15 \mu\text{m}$ without development of cracks or the presence of large voids. Densification of the coatings was carried out as a post-EPD process and to improve the adhesion of the coatings to the substrate. The sintering temperature was 340°C , sintering time 20 min and heating rate 300°C h^{-1} (Boccaccini *et al.* 2006c). PEEK, a biocompatible and stable

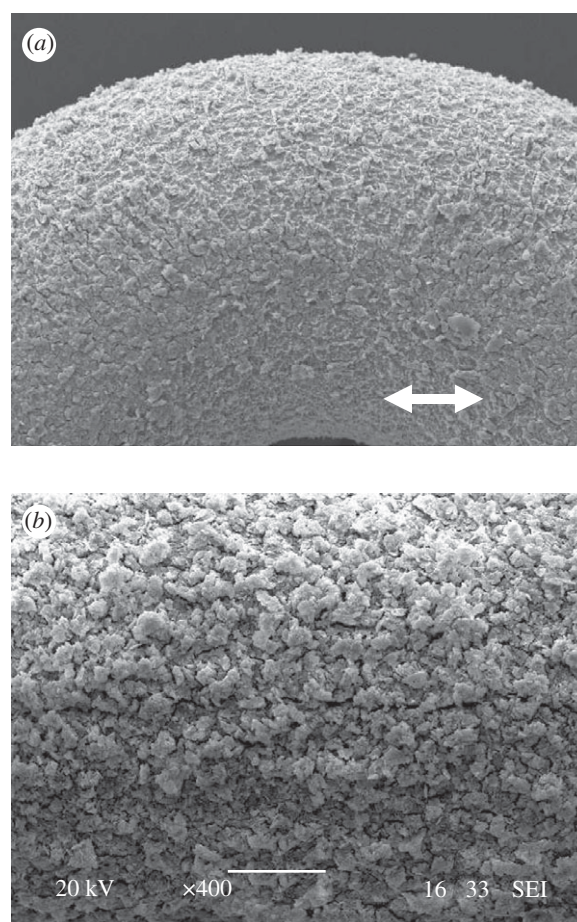


Figure 5. SEM images of PEEK/Bioglass-coated NiTi wires at (a) low and (b) high magnification showing the high uniformity of the coating structure achieved. (EPD parameters: Voltage = 20 V, time = 5 min, PEEK concentration = 6 wt%, Bioglass concentration = 1 wt%) (Boccaccini *et al.* 2006c). Scale bars, (a), (b) $50 \mu\text{m}$.

polymer that has been deposited by EPD (Wang *et al.* 2003; Corni *et al.* 2009), is suitable for several medical device applications since it combines outstanding chemical and hydrolysis resistance, high strength and excellent tribological properties. To impart bioactivity to PEEK coatings, Bioglass particles have been incorporated to the coating (Boccaccini *et al.* 2006c). Figure 5 shows scanning electron microscope (SEM) images at different magnification of a PEEK/Bioglass-coated NiTi wire obtained by EPD (20 V, time = 5 min, PEEK concentration in ethanol = 6 wt%, Bioglass concentration = 1 wt%). The coatings were seen to exhibit excellent adherence to the substrate and high microstructural homogeneity (Boccaccini *et al.* 2006c).

3. POROUS AND COMPOSITE BIOMATERIALS

Beyond the application of EPD to fabricate free-standing ceramic components for biomedical applications, e.g. alumina acetabular cup for hip replacement or piezoelectric microactuators for minimally invasive surgery procedures (Ma *et al.* 2007), it is interesting to consider the novel application of the technique in the

development of porous and composite biomaterials, where EPD has substantial advantages in comparison with conventional fabrication methods.

3.1. Porous materials

EPD has been increasingly used to coat textile and porous substrates with bioactive ceramic particles to produce a range of porous materials that can be applied for bioactive scaffolds, drug delivery systems and orthopaedic implants. Zhitomirsky (1998) demonstrated that HA-coated carbon fibres can produce, after burning out the fibrous carbon substrates, hollow HA fibres of various diameters. A similar study was carried out by Wang *et al.* (2002), who performed repeated HA deposition on carbon rods in order to obtain a thick, uniform and crack-free HA film. After burning out the carbon rod a uniform and crack-free HA ceramic tube could be produced. EPD has also been applied by Ma *et al.* (2003b) to prepare bioactive porous HA scaffolds. They showed that the pores were interconnected and that pore size was between several micrometres and hundreds of micrometres, as desired for tissue engineering applications (Karageorgiou & Kaplan 2005). Moreover, these scaffolds demonstrated excellent mechanical properties.

The deposition of bioactive glass particles onto three-dimensional porous biopolymer scaffolds to impart bioactivity for bone tissue engineering was introduced for the first time in 2002 (Roether *et al.* 2002). In these cases, when the substrate to be coated is not electrically conductive, the term electrophoretic impregnation can be applied, as strictly speaking EPD is not occurring (Boccaccini & Zhitomirsky 2002). Related developments of electrophoretic impregnation of porous biomaterials materials have been carried out by Yamaguchi *et al.* (2009) and Yabutsuka *et al.* (2006), who deposited wollastonite particles (less than 1 μm in size) into the pores of porous alumina and porous ultrahigh-molecular-weight polyethylene (UHMWPE). Similar wollastonite coatings on porous alumina were developed by Ozawa *et al.* (2003). The suspension for EPD was based on acetone with added iodine. These composites were soaked in SBF and it was found that apatite crystals were formed from the wollastonite particles growing on the surfaces and covering the entire porous composite. The bonding strength of the apatite layer to the substrates was as high as 8.9 MPa for alumina and 5.2 MPa for UHMWPE owing to an interlocking effect (Yamaguchi *et al.* 2009). Bioactive glass coatings on biomorphic SiC have also been produced by EPD (Ria *et al.* 2008). A post-deposition thermal treatment was carried out to improve the properties of the coatings. The analysis demonstrated that the electrophoresis parameters, such as voltage, distance between electrodes and deposition time, play an important role in determining the thickness and micro-structural homogeneity of the coatings. The post-deposition thermal treatment produced the cohesion of glass particles, leading to a uniform coating with excellent coverage of the porous SiC surface morphology. Given the increased interest in the use of SiC porous structures in the biomedical field, it is expected that

the bioactivation of the three-dimensional surfaces of porous bodies or textiles (SiC fibre performs) by EPD will gain increased research attention.

The direct application of EPD to fabricate porous ceramics has been demonstrated using alumina powder-stabilized Pickering emulsions of paraffin oil in ethanol (Neirinck *et al.* 2009a). The pore-forming paraffin is extracted from the consolidated powder compact by means of evaporation prior to sintering. The final product contains spherical pores with diameters that can be tuned from 200 to 20 μm . Both open and closed porosities can be obtained by altering the emulsion composition. The method is also promising for use with other bioceramics.

3.2. EPD of biocomposite materials

The development of fibre-reinforced ceramic matrix composites by EPD is well established, as reviewed elsewhere (Boccaccini *et al.* 2001). It is highly possible to adapt the technique for the fabrication of structural bioceramic composites of possible relevance in the biomedical field, for example with silica, zirconia or alumina matrices. However, to the authors' knowledge, only limited work has been carried out using bioactive ceramic matrices. One of such few studies is the work of Ordnung *et al.* (2005) on the use of EPD to fabricate HA matrix composites reinforced by alumina fibre fabrics. Indeed, further investigations in the field would be relevant if the intrinsic low fracture toughness of HA and bioactive glasses should be improved by fibre reinforcement for load-bearing medical applications.

Further efforts have been devoted to the development of EPD for fabricating laminated and functionally graded ceramic composites, in particular in the system zirconia/alumina, owing to the high fracture resistance of these structures and their potential use in biomedical implants (Kaya 2003).

Functionally graded $\text{Al}_2\text{O}_3/\text{ZrO}_2$ materials have also been developed by EPD for ball heads and acetabular cup inserts (Anné *et al.* 2006). A composition gradient in alumina and zirconia was realized to obtain a pure alumina surface region and a homogeneous alumina/zirconia core with intermediate continuously graded regions. The experimental set-up for graded $\text{Al}_2\text{O}_3/\text{ZrO}_2$ FGM ball heads has been described by Anné *et al.* (2006), and it is shown in figure 6. The rig is composed of a deposition cell, a suspension circulation system driven by a peristaltic pump, a mixing cell where a pure alumina suspension can be mixed with the $\text{Al}_2\text{O}_3\text{--ZrO}_2$ mixed suspension using a magnetic stirrer and a controlled suspension supply system to add the $\text{Al}_2\text{O}_3\text{--ZrO}_2$ suspension into the mixing cell. The deposition cell consists of a deposition electrode on which the powder compact is formed, a counter electrode, a positioning device and a suspension flow through the system. The deposition electrode has the shape of the outer side of the ball head, and it consists of an upper and lower part, to allow the removal of the EPD deposit.

Based on the promising results achieved so far, a significant growth of R&D work related to the EPD of laminated and functionally graded bioceramics for

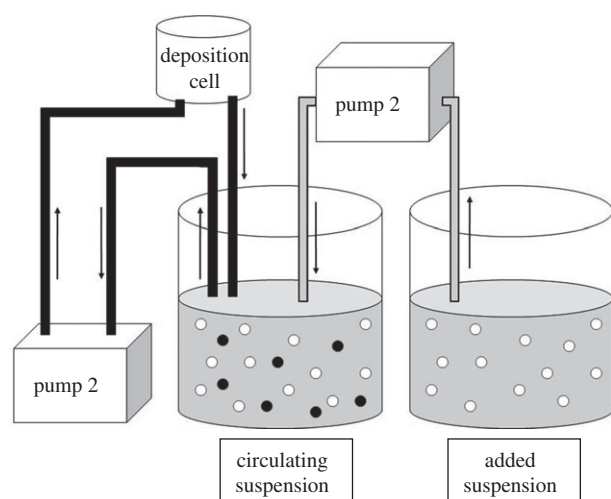


Figure 6. Schematic diagram of the experimental set-up used to fabricate complex-shaped functionally graded alumina and zirconia-based femoral ball heads by EPD (Anné *et al.* 2006).

medical applications can be anticipated, for example for stronger and tougher structural ceramic and composite components for orthopaedic (structural) applications. Moreover, EPD fabrication of bioactive ceramic composites based on HA or bioactive glass matrices reinforced by suitable biocompatible fibres remains a challenge for future research efforts.

3.3. Biomimetic coatings

EPD represents a suitable method for the development of a new family of organic–inorganic biomimetic coatings. These coatings are developed to mimic nacre and other natural structures characterized by a unique ordered ‘brick and mortar’ microstructure (Ruiz-Hitzky *et al.* 2005; Luz & Mano 2009). Lin *et al.* (2008c) have shown the combination of hydrothermal processing and EPD to prepare nacre-like coatings, where a two-step assembly process involved the intercalation of the polymer phase into the interlayer space of montmorillonite and subsequent EPD on a metallic substrate. The experimental set-up is shown schematically in figure 7.

The polymer used was a type of acrylic resin. Issues related to possible toxicity as a consequence of the use of this particular resin were not reported (Lin *et al.* 2008c). The completely polymerized resin should not lead to any irritations or allergies, but the polymerization process of acrylic resins may result in incomplete polymerization of all monomers. Thus, the residual monomers can cause cytotoxicity (Jorge *et al.* 2003). Although the biomedical application of these biomimetic coatings has not been investigated so far, there is huge potential for the further development of EPD in the field of biomimetic coatings involving non-toxic materials. Long *et al.* (2007) developed polyacrylamide–clay nacre-like nanocomposites by EPD from aqueous suspensions. In this approach, Na-montmorillonite was dispersed in deionized water and vigorously stirred for 3 h; subsequently, an acrylamide aqueous solution containing HCl to adjust the pH was added slowly

to prepare an organic clay. This clay was then used for EPD employing stainless steel plates as electrodes separated by 2 cm and a voltage of 5 V. The well-ordered layer-by-layer platelet-like structure obtained exhibited relatively high hardness and Young’s modulus. Again, the biomedical application was not reported, which could be compromised considering that acrylamide is a monomer that should have a carcinogenic and genotoxic impact on the human body. After complete polymerization to polyacrylamide, however, there should be no toxic effects (Lange-Ventker 2001). In a related investigation, Lin *et al.* (2009) developed EPD for assembling gibbsite nanoplatelets into self-standing films. The positively charged gibbsite platelets were dispersed in a water–ethanol mixture and deposited using a parallel-plate sandwich EPD cell consisting of an indium tin oxide (ITO) working electrode, a gold counter electrode and a polydimethylsiloxane spacer. The electric field used was 1100 V m^{-1} . Under these conditions, nanoplatelets can be aligned preferentially in parallel to the electrode surface. After drying, the gibbsite films on the ITO electrode could be peeled off by using a razor blade. To form biomimetic composites, the interstices between the assembled nanoplatelets were infiltrated with polymer (e.g. acrylate-based resin), leading to optically transparent films. The assembly of gibbsite nanoplatelet/polyvinyl alcohol nanocomposites by EPD in a single step has also been demonstrated to produce biomimetic composites (Lin *et al.* 2008b). The approach can be extended to include other materials of higher relevance for the biomedical field, for example bioactive glass flakes and biodegradable polymers to form bioactive nanocomposites. It is worthwhile noticing that EPD provides in this regard an alternative to other bottom-up approaches (e.g. LBL assembly) (Luz & Mano 2009).

4. APPLICATIONS IN NANOBIO TECHNOLOGY

4.1. EPD of nanoparticles

The EPD of ceramic nanoparticles (size less than 100 nm) is a special case of colloidal processing being applied to produce a variety of materials, including biomaterials of high microstructural homogeneity. Previous work on EPD of nanoparticles has been reviewed comprehensively elsewhere (Corni *et al.* 2008) and only selected recent reports of relevance for the field of biomaterials are covered in this section.

Besides the research discussed above on EPD of HA nanoparticles (see also table 1), other ceramic nanoparticles with potential use in the biomedical field have been considered. Calcium phosphate nanoparticles, for example, were electrophoretically deposited on silicon substrates, which were further machined by laser direct writing to obtain guiding structures (grooves of 0.5 mm in depth and 4–20 mm in width) for osteoblast cell alignment (Wieman *et al.* 2007). Similar structures have also been created on titanium substrates (Urch *et al.* 2006).

Silica nanoparticles and mesoporous silica layers are relevant for a number of biomedical applications (Cousins *et al.* 2004; Vallet Reggi *et al.* 2006).

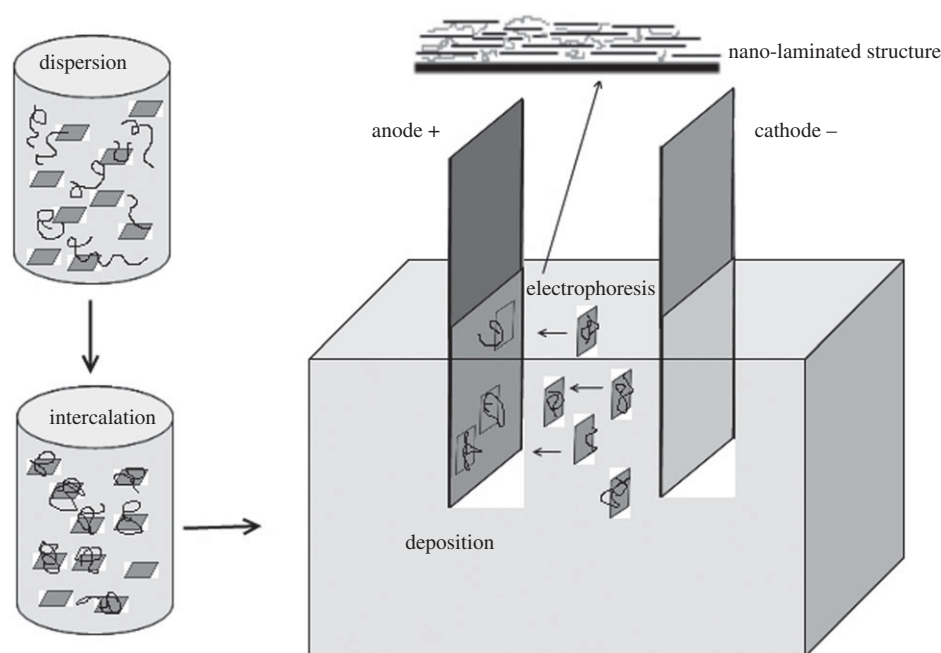


Figure 7. Experimental set-up developed by Lin *et al.* (2008c) to prepare nacre-like coatings, where a two-step assembly process is applied, involving the intercalation of the polymer phase into the interlayer space of montmorillonite and subsequent EPD on a metallic substrate.

Tabellion & Clasen (2004) have discussed previous work on the fabrication of large components, free-standing objects, hollow bodies and objects of complex three-dimensional shape using EPD of silica nanoparticles in aqueous suspensions. It has been proposed that silica nanoparticulate coatings can be useful model materials to develop reproducible surface topography to influence cellular response (Cousins *et al.* 2004) and EPD is thus a convenient technique to order silica nanoparticles in predetermined patterns. Nishimori and co-workers (Nishimori *et al.* 1996; Hasegawa *et al.* 1998) have developed an EPD-based sol-gel technique to deposit thick silica films on stainless steel substrates. More recently, silica nanoparticles (size 350 nm) were deposited by EPD on stainless steel substrates, leading to uniform silica coatings of about 80 μm in thickness (Tada & Frankel 2007). Finer silica nanoparticles (approx. 10 nm) suspended in isopropanol have also been deposited by EPD on nanostructured copper electrodes to form sensors (Bazin *et al.* 2008).

The deposition of alumina nanoparticles on metallic substrates has also been reported (Clark *et al.* 1988). In this case, however, no specific biological characterization of the coatings has been carried out. The production of nanostructured zirconia coatings by EPD from aqueous suspensions has been reported for the fabrication of dental crowns (Moritz *et al.* 2006; Oetzel & Clasen 2006). In a report by Jin *et al.* (2008), a high solid content (greater than 75 wt%) suspension of biocompatible yttria-stabilized tetragonal zirconia powder (mean particle size 40 nm) in distilled water was used. The ceramic dental restoration framework was fabricated using a porous gypsum model as cathode, which was previously soaked in electrical conductive liquid to impart electrical conductivity.

The deposition of nanostructured titania films by EPD has been carried out by several groups

(Lin *et al.* 2006; Ofir *et al.* 2006; Tang *et al.* 2006; Moskalewicz *et al.* 2007; Santillián *et al.* 2008). These coatings can find applications in the orthopaedic field and for antibacterial applications. In most cases, TiO_2 nanoparticles suspended in acetylacetone with addition of iodine are used and EPD is carried out under constant voltage conditions (10–20 V) for deposition times less than 10 min, leading to a high degree of particle packing in homogeneous film microstructures of up to 20 μm in thickness (Boccaccini *et al.* 2004; Moskalewicz *et al.* 2007). A high-temperature sintering treatment is required to densify the porous electrophoretic TiO_2 deposits. In the investigation of Moskalewicz *et al.* (2007), nanocrystalline TiO_2 powders were electrophoretically deposited on the surface of Ti–6Al–7Nb alloy substrate. Post-heat treatment at 850°C was performed to improve the adhesion strength of the coating. Microstructural analyses showed that the as-deposited nanocrystalline TiO_2 films exhibited both anatase and rutile phases. The electrophoretic codeposition of microsized PS beads combined with nanosized TiO_2 was shown recently by Radice *et al.* (2010) with the aim of preparing biomedical coatings exhibiting micropatterned surfaces combined with nanotopography, where the micropatterning is determined by the microsized template (PS spheres are burnt out during sintering) and the nanotopography is provided by the sintered titania nanoparticles. Such surface structuring is relevant for biomedical applications, where specific combinations of micro- and nanoroughness are confirmed to control cell adhesion and osteointegration of implant surfaces (Zhao *et al.* 2006b), which was also discussed above. The electrophoretic fabrication of such coatings with scale-resolved structuring requires well-controlled particle functionalization, which is still a challenge, as discussed by Radice *et al.* (2008) investigating PS– TiO_2 composite particle

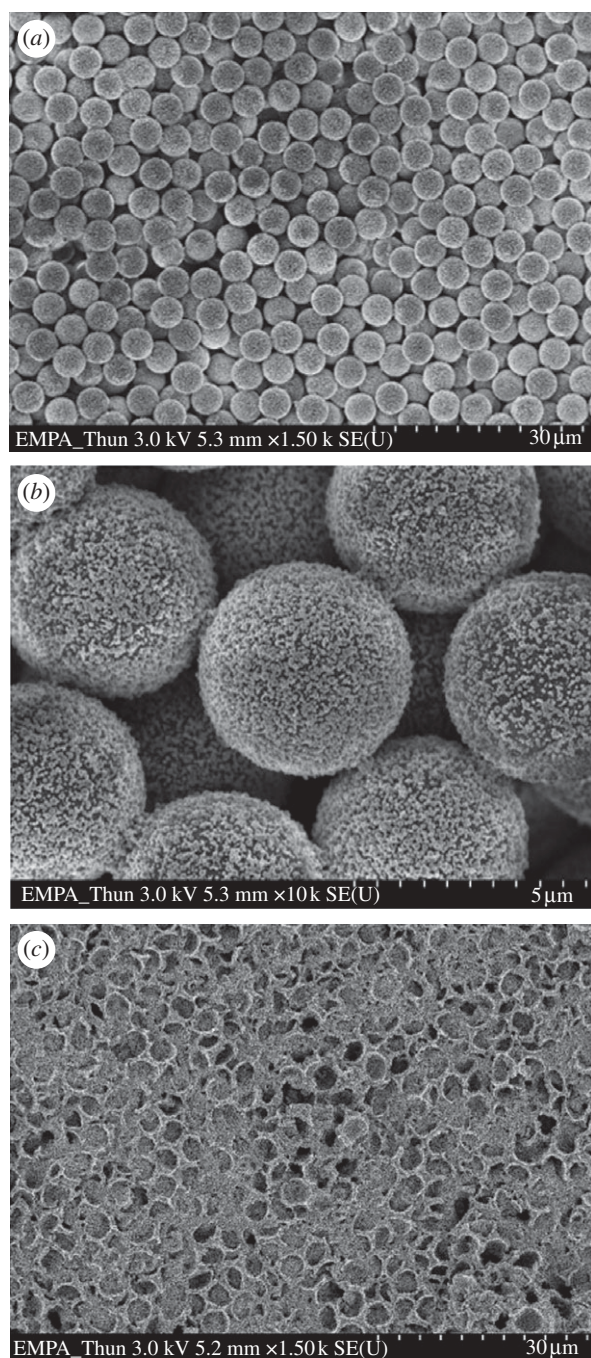


Figure 8. SEM images of PS–TiO₂ electrophoretic deposits, at different magnifications, (a,b) before and (c) after sintering, demonstrating the successful development of a scaled-structured titania coating for orthopaedic applications (Radice *et al.* 2008, 2010). (Images are courtesy of Dr S. Radice.)

suspensions in isopropanol. Sintering was carried out up to a maximal temperature of 900°C, holding for 2 h in argon. Figure 8 shows SEM images of the PS–TiO₂ electrophoretic deposits, at different magnifications, before and after sintering, demonstrating the successful development of a scaled-structured titania coating of relevance for orthopaedic applications (Radice *et al.* 2008).

There is increasing interest in developing EPD of TiO₂ nanoparticles from aqueous suspensions for biomedical applications (Lebrette *et al.* 2006). Moreover Ag-doped TiO₂ nanoparticles are being considered to

develop anti-infectious coatings by EPD (Santillán *et al.* in preparation). EPD has been used also to fabricate high-conductivity TiO₂/Ag fibrous structures by EPD (Hosseini *et al.* 2008).

Given that TiO₂ nanotubes are of high relevance for studies of cellular compatibility of nanostructures (Bauer *et al.* 2008), the EPD of titania nanotubes (Wang *et al.* 2006a) becomes relevant for advanced applications of titania nanostructures in the biomedical field, for example for fundamental investigations of the effect of nanotopography on cell behaviour (Park *et al.* 2007). Combinations of TiO₂ nanoparticles and CNTs developed by EPD are described in §5.5. To conclude this section, it should be mentioned that a series of other coatings, notably TiN coatings of possible relevance in the field of medical devices, have also been fabricated by EPD (Cui *et al.* 2009). Similarly, EPD has been applied to deposit diamond nanoparticles from isopropyl alcohol suspensions on graphite substrates (Affoune *et al.* 2001). However, a further analysis of these electrophoretic coating systems, which, to the authors' knowledge, have not been yet developed specifically for biomedical applications, is beyond the scope of the present review paper.

4.2. Organic–inorganic and organic nanocomposites

As mentioned above, sintering of bioceramic coatings on metallic substrates at high temperatures can result in the degradation of the metallic substrates. There is thus a clear need to develop bioactive coatings that can be consolidated and attached to metallic substrates at room temperature. Moreover, in the case of HA coatings for orthopaedic applications, it is obvious that the microstructure of sintered HA is different from that of nanostructured HA in bone tissue. The unique mechanical and functional properties of bone result from its specific multi-layer composite structure, composed of collagen nanofibre layers, which act as reinforcement, and HA nanocrystals (Zhang *et al.* 2003a; Fratzl *et al.* 2004). Thus, there is increasing interest in developing organic–inorganic composites for biomedical applications, in the case of orthopaedics and bone regeneration strategies, and the goal is to mimic the structure of bone by combining nanoscale inorganic and biopolymer phases. In the case of coatings, a significant advantage of using a polymer as binder is the ability to adhere the coating strongly to the metallic substrate, avoiding the high-temperature densification step required for pure ceramic coatings (Ramaswamy *et al.* 2009).

Besides the use of synthetic stable (i.e. non-biodegradable) polymers, such as PEEK (Wang *et al.* 2003), to fabricate bioactive composite coatings by EPD (Boccaccini *et al.* 2006c; discussed in §2.4), a wide range of novel functional nanocomposite coatings incorporating natural or synthetic biodegradable polymers and nanosized fillers with or without the addition of bioactive molecules is being developed by adaptation of the EPD technique, which will be discussed in detail in this section.

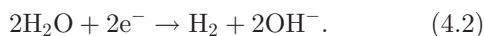
An important family of functional organic–inorganic coatings being developed currently for biomedical applications involve polysaccharides (Pang & Zhitomirsky 2005a,b; Cheong & Zhitomirsky 2008; Sun & Zhitomirsky 2009). Although other natural polymers such as lignin have been considered to develop HA/biopolymer coatings by EPD (Erakovic *et al.* 2009), the work on polysaccharides has been much more extensive, and it will be discussed in detail here. The interest in natural polysaccharides and inorganic nanoparticles stems from the recent discovery that polysaccharides form interfaces between organic and inorganic components in bone and govern the crystallization of HA nanoparticles (Wise *et al.* 2007). In this context, several recent studies have shown the possibility of EPD of organic–inorganic nanocomposites formed by HA nanoparticles in natural polysaccharide matrices, such as chitosan (CHIT) (Pang & Zhitomirsky 2005a,b), alginic acid (AlgH) (Cheong & Zhitomirsky 2008) and hyaluronic acid (HYH) (Sun & Zhitomirsky 2009).

4.2.1. Chitosan-based nanocomposites. CHIT is a natural cationic polysaccharide that can be produced by alkaline N-deacetylation of chitin. Important properties of this material, such as antimicrobial activity, chemical stability, biocompatibility, advanced mechanical and other properties, have been used in biomedical implants, scaffolds, drug delivery systems, biosensors and other biomedical devices (Sinha *et al.* 2005). The interest in CHIT for the fabrication of composite coatings by EPD stems from the cationic nature of this polymer in aqueous solutions and its excellent film-forming properties (Simchi *et al.* 2009).

Water soluble and positively charged CHIT (Wu *et al.* 2002; Zhitomirsky 2006) can be prepared by protonation in acidic solutions:



It is known that the increase in pH results in precipitation of CHIT at pH 6.6. In aqueous solutions, the cathodic reduction of water results in increasing pH at the cathode surface:



Therefore, the deposition of CHIT is achieved via the electrophoretic motion of cationic CHIT to the cathode and the neutralization of the positively charged CHIT macromolecules in the high pH region at the cathode surface (Simchi *et al.* 2009). The deposition process results in the formation of insoluble CHIT films:



HA–CHIT coatings were prepared by EPD from CHIT solutions containing dispersed HA nanoparticles (table 2). The CHIT content in the deposits was varied in the range 0–100% by variation of CHIT and HA concentrations in the deposition bath (Pang & Zhitomirsky 2005a,b, 2007). This approach offers the advantage of high-deposition rate and the possibility of formation of uniform coatings of controlled thickness on substrates of complex shape. The film thickness was varied in the range 0–200 μm and the coatings were

shown to provide corrosion protection of metallic substrates in SBF solutions. An important finding was the possibility of fabrication of nanocomposite films with preferred orientation of HA nanoparticles (Pang & Zhitomirsky 2005a,b, 2007). The crystallographic *c*-axis of the needle-shaped HA nanoparticles was oriented parallel to the substrate surface. It is important to note that similar preferred orientation of HA nanoparticles has been observed in natural bone (Zhang *et al.* 2003a; Fratzl *et al.* 2004). Therefore, the ability to control the orientation of elongated nanoparticles is important for the fabrication of advanced bone-substitute materials considering that the anisotropic orientation of HA nanocrystals plays an important role in the mechanical and other properties of natural bone (Fratzl *et al.* 2004). Further investigations have shown that the orientation of HA in nanocomposites prepared by EPD can be mainly attributed to interactions between HA and CHIT. The decrease in CHIT concentration in the deposits resulted in reduced orientation of the HA nanoparticles.

Several investigations have focused on the kinetics of deposition and deposition mechanism of CHIT-based nanocomposites (Pang & Zhitomirsky 2005a,b; Pang *et al.* 2009; Zhitomirsky *et al.* 2009). It was shown, for example, that CHIT molecules adsorbed on HA nanoparticles provided efficient electrosteric stabilization of HA in the suspensions. Moreover, adsorbed CHIT imparted a pH-dependent positive charge required for cathodic EPD. Deposition yield measurements showed a nonlinear increase in deposition yield with increasing particle concentration. It should be noted that the Hamaker equation (Hamaker 1940) predicts a linear increase in deposit mass M with increasing particle concentration C_s in dilute suspensions:

$$M = \mu EtSC_s, \quad (4.4)$$

where μ is the particle mobility in an electric field E , t is the deposition time and S is the electrode area. Of relevance for the present discussion, a moving boundary model for two-component systems has been developed in order to explain the nonlinear increase in the EPD yield from CHIT–HA suspensions (Pang *et al.* 2009). Taking into account that HA and CHIT form a composite deposit and a common deposit–suspension boundary, the deposition yield can be given by the following equation:

$$M = \mu EtSC_s \left(1 - \frac{C_s}{C_c}\right)^{-1} + \mu' EtS\psi_s \left(1 - \frac{\psi_s}{\psi_c}\right)^{-1}, \quad (4.5)$$

where C_c is the particle concentration in the deposit, μ' is the mobility of CHIT macromolecules, ψ_s is the concentration of CHIT in suspension and ψ_c is the concentration of CHIT in the deposit. The experimental data showed the increase in C_s/C_c and ψ_s/ψ_c with increasing C_s and explained the nonlinear dependence of the deposition yield in agreement with the moving boundary model.

The use of CHIT enables room-temperature processing of the composite coatings of controlled composition and microstructure, eliminating the problems related to the high-temperature sintering of HA

Table 2. Experimental conditions for EPD of organic–inorganic nanocomposites based on chitosan, alginate acid and hyaluronic acid.

coating composition	deposition conditions			references
	bath composition	solvent	electric field or current density	
HA–CHIT	0–8 gl ^{−1} HA	ethanol (83%)–	0.1 mAcm ^{−2}	Pang & Zhitomirsky (2005a,b, 2007)
	0.1–1 gl ^{−1} CHIT	water (17%)	10–40 Vcm ^{−1}	
HA–CHIT–CNT	0–5 gl ^{−1} HA	ethanol (83%)–	7–30 Vcm ^{−1}	Casagrande <i>et al.</i> (2008); Grandfield <i>et al.</i> (2009)
	0.5 gl ^{−1} CHIT	water (17%)		
	0–0.2 gl ^{−1} CNT			
HA–Silica–CHIT	0–5 gl ^{−1} HA	ethanol (83%)–	20 Vcm ^{−1}	Grandfield & Zhitomirsky (2008)
	0–3.6 gl ^{−1} SiO ₂	water (17%)		
	0.5 gl ^{−1} CHIT			
HA–CaSiO ₃ –CHIT	0–4 gl ^{−1} HA	ethanol (83%)–	15 Vcm ^{−1}	Pang <i>et al.</i> (2009)
	0.5 gl ^{−1} CHIT	water (17%)		
	0.1–0.5 gl ^{−1} CaSiO ₃			
Apatite–wollastonite/CHIT laminates	2 gl ^{−1} AW	ethanol or	1(CHIT) or	Sharma <i>et al.</i> (2009a,b)
	0.2 gl ^{−1} CHIT	water	3 (AW) mAcm ^{−2}	
HA–CHIT/CHIT laminates	0–4 gl ^{−1} HA	ethanol (83%)–	15 Vcm ^{−1}	Pang <i>et al.</i> (2009); Sun <i>et al.</i> (2008)
	0.5 gl ^{−1} CHIT	water (17%)		
HA–CHIT/Ag–CHIT laminates	0–3 gl ^{−1} HA	ethanol (83%)–	0.1 mAcm ^{−2}	Pang & Zhitomirsky (2008)
	0–1 mM AgNO ₃	water (17%)		
	0.5 gl ^{−1} CHIT			
CHIT–HA–bioglass	0–1 gl ^{−1} HA	ethanol (83%)–	15 Vcm ^{−1}	Sun <i>et al.</i> (2008)
	0–0.5 gl ^{−1} bioglass	water (17%)		
	0.5 gl ^{−1} CHIT			
AlgH–HA	1–2 gl ^{−1} AlgNa	ethanol (60%)–	1–2 mAcm ^{−2}	Cheong & Zhitomirsky (2008)
	2–6 gl ^{−1} HA	water (40%)		
AlgH–TiO ₂	1–2 gl ^{−1} AlgNa	ethanol (60%)–	1–2 mAcm ^{−2}	Cheong & Zhitomirsky (2008)
	2–6 gl ^{−1} HA	water (40%)		
AlgCa–CaCO ₃ –bacterial cells	9 gl ^{−1} AlgNa	water	0.3 mAcm ^{−2}	Shi <i>et al.</i> (2009)
	2.3 gl ^{−1} CaCO ₃			
HYH–HA	0.5–1 gl ^{−1} HYN	ethanol (70%)–	20 Vcm ^{−1}	Sun & Zhitomirsky (2009)
	0.5–4 gl ^{−1} HA	water (30%)		
HYH–TiO ₂	0.5–1.0 gl ^{−1} HYN	ethanol (70%)–	10–20 Vcm ^{−1}	Ma <i>et al.</i> (2010b)
	1–2 gl ^{−1} TiO ₂	water (30%)		

(Pang & Zhitomirsky 2005a,b, 2007; Simchi *et al.* 2009). Moreover, the use of CHIT has enabled the EPD fabrication of a new family of novel nanocomposite coatings containing other functional materials as filler. Composite CHIT–silica films were deposited for the fabrication of immunosensors (Liang *et al.* 2008) and implants (Grandfield & Zhitomirsky 2008). Films were obtained with a three-dimensional porous structure, which possessed a high surface area, good mechanical stability and good hydrophilicity (Liang *et al.* 2008). Such films provide a biocompatible microenvironment for maintaining the bioactivity of immobilized proteins and enable protein loading. Many investigations have focused on the fabrication of CHIT–CNT and CHIT–CNT–HA composites, containing pristine CNT (Grandfield *et al.* 2009), as well as CNT containing cationic and anionic functional groups (Casagrande *et al.* 2008). In all cases, composite materials containing CNT showed improved mechanical properties (Jia *et al.* 2009). In related research, composite CHIT–CNT films were developed for the fabrication of efficient impedance cell sensors (Hao *et al.* 2007). CHIT–CNT films were functionalized with the enzyme glucose oxidase (Zhou *et al.* 2007; Meyer *et al.* 2009; Zeng *et al.*

2009) for specific biosensor applications. The biosensors exhibited high reproducibility, long-time storage stability and good anti-interference ability. In addition, EPD has been used for the fabrication of CHIT–CNT–Nile Blue–horseradish peroxidase composite films (Xi *et al.* 2009). These composites are interesting for biosensors as they show high electrocatalytic activity and fast amperometric response to H₂O₂.

In a set of recent publications, various methods have been developed for the immobilization of proteins, nucleic acids and virus particles (Yi *et al.* 2005; Yang *et al.* 2009) in the CHIT matrix, and the microfabrication of novel devices for biomedical applications has been investigated. Bovine serum albumin (BSA), for example, was used as a model protein for the development of electrochemistry-based approaches to incorporate proteins into CHIT films (Ma & Zhitomirsky 2010). Figure 9a shows a typical SEM image of a CHIT–BSA film recently developed by Ma & Zhitomirsky (2010). EPD was confirmed to be a suitable method for the deposition of these protein–polymer thick films of controlled thickness and it was confirmed that the excellent film-forming and binding properties of CHIT enabled the formation of good-

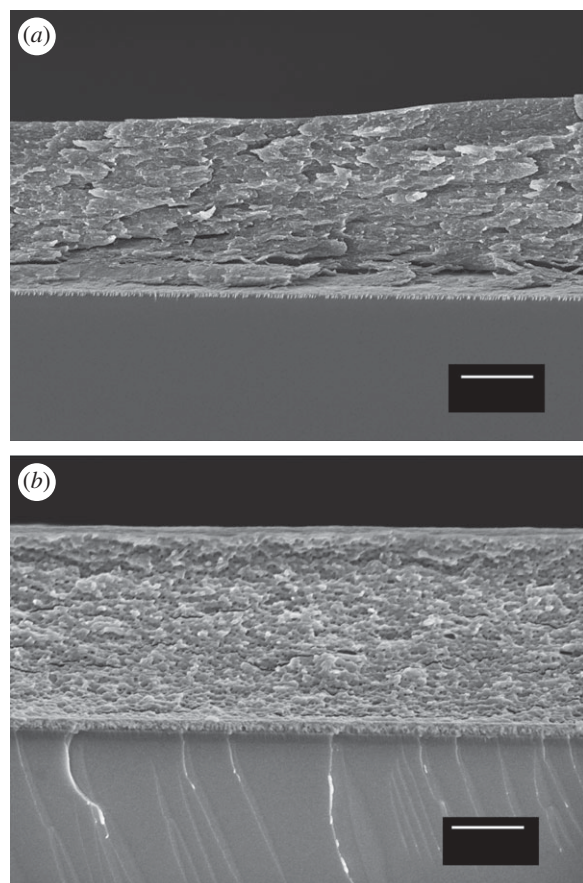


Figure 9. SEM images of cross sections of (a) CHIT-BSA and (b) alginate-BSA composite films confirming the electrophoretic formation of relatively uniform films of controlled thickness (Ma & Zhitomirsky 2010; Sun & Zhitomirsky 2010). (Reproduced with permission of Maney Publishing and the Editors of *Surface Engineering*). Scale bars, (a) 3 μm ; (b) 1 μm .

quality, adherent films. Composite films containing MnO_2 , Fe_3O_4 and ZnO nanoparticles in the CHIT matrix have also been prepared by EPD and proposed for applications in biosensors (Zhao *et al.* 2006a; Ling *et al.* 2009; Li *et al.* 2010a). Moreover, three-phase composites incorporating HA nanoparticles and Bioglass particles in CHIT matrices have also been developed by EPD (Zhitomirsky *et al.* 2009).

In related experiments, heparin has been investigated as a model drug for the fabrication of composite coatings with potential for drug delivery (Sun *et al.* 2008, 2009). In this strategy, the non-stoichiometric complexes of anionic heparin and cationic CHIT were used for cathodic deposition of composite films. The ability of the CHIT-heparin films to bind antithrombin, as measured by binding of ^{125}I -radiolabelled antithrombin, was much greater than that for unmodified CHIT films, suggesting that EPD can be used for the surface modification of Nitinol and other alloys with functional CHIT-heparin coatings for improved haemocompatibility (Sun *et al.* 2008, 2009).

The development of advanced nanocomposites includes not only the selection of components but also their careful structural design. It is well known, for example, that the unique performance of natural tissues such as bone or dentine arises from the precise hierarchical organization, graded composition, localized

porosity and relative orientation of HA nanocrystals and collagen (Fratzl *et al.* 2004). The investigation of the hierarchical structure of natural bone has stimulated further expansion of the applications of EPD in the biomedical field. As mentioned above, EPD is ideally suited for the fabrication of multi-layer and FGM composites. In the case of polymer-bioceramic nanocomposites, EPD has been developed for the fabrication of multi-layer and FGM films containing layers of pure CHIT, CHIT-HA, CHIT-Ag, CHIT-CNT and CHIT-silica (Grandfield & Zhitomirsky 2008; Pang & Zhitomirsky 2008; Sun *et al.* 2008; Pang *et al.* 2009). It was shown in those studies that the thickness of the individual layers in a multi-layer of functionally graded coating system can be controlled by variation of EPD parameters. Such multi-layer structures are of interest in a range of applications, where different fillers confer different functionalities or capabilities to the organic films. For example, in CHIT-Ag nanocomposites, the controlled Ag release from the composite coatings open opportunities for their use as antibacterial coatings (Pang & Zhitomirsky 2008), with the interfacial CHIT layer providing improved adhesion of the composite films to the metallic substrates. In related studies, apatite-wollastonite (AW)/CHIT deposits have been prepared by constant voltage EPD on Ti substrates (Sharma *et al.* 2009a,b) for orthopaedic applications. It was found that solution pH and current density are important factors controlling deposition yield, coating microstructure, composition and properties. The coating exhibited a porous structure with interconnected porosity and ceramic particles entrapped inside the polymer layers. The addition of CHIT increased the adhesive strength of the composite coating, and the Young's modulus of the coating was found to be 9.23 GPa. Bioactivity studies showed that CHIT enhanced the apatite formation rate in SBF, and apatite particles formed in SBF showed sheet-like morphology. It was also shown that AW/CHIT coatings exhibited good adhesion to Ti substrates and haemocompatibility (Sharma *et al.* 2009a,b), as well as faster bone healing when implanted in rabbit tibiae (Sharma *et al.* 2009c).

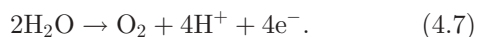
4.2.2. Alginate-based nanocomposites. Recent studies have confirmed the feasibility of EPD to develop AlgH and AlgH-HA nanocomposites (Cheong & Zhitomirsky 2008). AlgH and alginates are natural anionic biodegradable, biocompatible, non-toxic and low-cost biopolymers, which are being used in a range of biomedical applications, such as wound dressing, bone and nervous tissue repair, drug delivery and for the encapsulation of cells and enzymes (Becker *et al.* 2001). Alginate, being an anionic polymer with carboxyl end groups, is an attractive material for the fabrication of composite coatings by EPD. Thin films of AlgH were obtained by anodic deposition on various conductive substrates (Cheong & Zhitomirsky 2008). The proposed mechanism of AlgH deposition is based on electrophoresis of anionic Alg^- macromolecules and charge neutralization of Alg^- in the low pH region at

the anode surface, as investigated by Cheong & Zhitomirsky (2008).

It has been proposed that the dissociation of sodium alginate (AlgNa) in aqueous solution (pH 9) results in the formation of anionic Alg^- species:



The electrochemical decomposition of water results in pH decrease at the anode surface:



It is known that alginate solutions form gels at pH less than 3. Therefore, the electrophoresis and neutralization of negatively charged Alg^- species in the acidic region at the anode surface results in the formation of AlgH deposits:



Further investigations have shown the possibility of formation of composite AlgH–HA films (Cheong & Zhitomirsky 2008). In this approach, the adsorption of anionic Alg^- species provided electrosteric stabilization of HA particles in suspensions. The anodic codeposition of negatively charged Alg^- and HA particles containing adsorbed Alg^- resulted in the formation of the composite films. The HA content was controlled by variation of the HA concentration in suspension and the film thickness was varied in the range 0–50 μm . EPD has been used also for the fabrication of CHIT/AlgH–HA laminates (Cheong & Zhitomirsky 2008). Moreover, the method allowed the fabrication of composite coatings containing other bioceramics (Cheong & Zhitomirsky 2008; Zhitomirsky et al. 2009). In another study, AlgH–CNT nanocomposite films were obtained on various conductive substrates (Grandfield et al. 2009), and it was shown that AlgH promotes the dispersion and deposition of CNTs. EPD of AlgH and horseradish peroxidase has been used for the application in biosensors (Liu et al. 2009). Anodic EPD has also been investigated for the fabrication of alginate-based composite coating containing drugs, such as heparin, and proteins, such as BSA (Ma & Zhitomirsky 2010; Sun & Zhitomirsky 2010). Figure 9b shows a typical SEM image of alginate–BSA composite film (Ma & Zhitomirsky 2010; Sun & Zhitomirsky 2010), confirming the formation of relatively uniform films of controlled thickness.

EPD has been recently used for the incorporation of bacterial cells into alginate films cross-linked with Ca^{2+} ions and containing CaCO_3 particles (Shi et al. 2009). The authors proposed a deposition mechanism, which is based on the use of 10 μm CaCO_3 particles, which were insoluble in the bulk of the solutions, but quickly dissolved at the electrode surface (Shi et al. 2009). The fast dissolution of the particles by electrogenerated low pH resulted in the release of Ca^{2+} and CO_2 . The film-formation mechanism was based on the cross-linking of Alg with Ca^{2+} ions and the formation of an insoluble AlgCa gel. The interesting feature of this approach is that efficient charge transfer, required for the electrode reactions, pH decrease and CaCO_3 dissolution, was achieved through thick polymer films with

thickness in the range of up to 1 mm. It was shown that the films prepared by this method can be used for cell-based biosensing (Shi et al. 2009).

4.2.3. Hyaluronic acid-based nanocomposites. HYH is another important natural polysaccharide for biomedical applications. HYH is presented at high concentrations in skin, joints and cornea. HYH is a bioactive material, associated with several cellular processes, including angiogenesis and the regulation of inflammation. In living tissue, HYH is usually combined with proteins to control various functions of the tissue. Anodic EPD has been developed for fabrication of HYH films and composites from sodium hyaluronate (HYNa) solutions in water or in a mixed ethanol–water solvent (Grandfield et al. 2009; Sun & Zhitomirsky 2009; Ma et al. 2010a,b). The deposition mechanism of this polysaccharide has been discussed in recent investigations (Sun & Zhitomirsky 2009; Ma et al. 2010a). It was suggested that the dissociation of HYNa results in the formation of anionic HY^- species:



The applied electric field provides electrophoretic motion of the anionic HY^- species towards the anode surface, where the pH decreases owing to the electrochemical decomposition of water (reaction (4.7)). It was suggested that the formation of HYH gel in the low pH region at the electrode surface can be achieved as a result of the charge compensation of $-\text{COO}^-$ groups:



However, no deposition of HYH was achieved from 0.5 g l^{-1} aqueous solutions of HYNa (Sun & Zhitomirsky 2009). In contrast, HYH films were obtained from 0.5 g l^{-1} HYNa solutions in a mixed ethanol–water solvent (Sun & Zhitomirsky 2009). The deposit formation was attributed to cross-linking of HYH in ethanol–water solutions in acidic conditions at the electrode surface. The cross-linking in acidic ethanol–water solutions occurs without change in the chemical structure and is related to the formation of hydrogen-bonding network among the chains. The cross-linking of HYH results in water-insoluble gels. However, recent investigations have shown that EPD of HYH can be achieved from aqueous HYNa solutions of higher concentrations (Ma et al. 2010a). The deposition rate increased with increasing HYNa concentration in solution. The film thickness was varied in the range 0–20 μm . The charge transfer through such thick insulating films presents difficulties. It is suggested that the surface pH decrease in reaction (4.7) is beneficial at the initial stage of deposition. The mechanism based on the surface pH decrease can explain the formation of thin or porous films. However, this mechanism cannot explain the formation of thick and dense films obtained from the concentrated solutions (Ma et al. 2010a). It is important to note that electrophoresis results in the accumulation of the charged polymer macromolecules at the electrode surface. It was suggested (Zhitomirsky 2002) that the increase in particle concentration can result in enhanced

depletion at the electrode surface, which promotes particle coagulation. Such interactions can result in attraction of similarly charged particles and explain the formation of thick films from the solutions of higher concentrations (Ma *et al.* 2010a).

Anodic EPD has been developed for the fabrication of composite films containing HA nanoparticles and CNTs in HYH matrices (Grandfield *et al.* 2009; Sun & Zhitomirsky 2009). It was found that HY^- species adsorbed on HA and CNTs provided efficient electrosteric stabilization and charging of the HA nanoparticles and CNTs. Film thickness was varied in the range 0–100 μm . The increase in HA nanoparticle concentration in suspensions resulted in increased HA content in the films. In another study, the kinetics of HYH–BSA deposition in aqueous solutions has been investigated (Ma *et al.* 2010a). It was shown that thick HYH–BSA films with thickness of up to 80 μm can be obtained from aqueous solutions. As in previous studies with CHIT and Alg, BSA was used as a model protein for the fabrication of composite films. Further investigations showed that other proteins and drugs can be incorporated into the HYH-based composite coatings (Ma *et al.* 2010b), which opens the possibility of fabrication of a range of novel composites containing other relevant drugs, proteins and enzymes in the HYH matrix.

4.2.4. Synthetic polymer-based nanocomposites. In comparison to the nanocomposite systems described above based on natural biopolymers, there has been much less research work on the use of EPD to develop nanocomposites with synthetic biopolymers. In one of the few investigations available, composite silicon-substituted HA-poly(ϵ -caprolactone) (HA/PCL) coatings have been prepared by Xiao *et al.* (2009). The use of PCL resulted in improved coating adhesion. After immersion in SBF for 8 days, HA/PCL coatings showed the ability to induce the formation of a bone-like apatite surface layer, which is the marker for bioactive behaviour as discussed above. The polymer content in the coating was limited to 5 wt% because the HA deposition rate decreased with increasing polymer concentration in the suspensions (Xiao *et al.* 2009).

5. CNT-CONTAINING BIOMATERIALS

5.1. CNTs as biomaterials

The remarkable high mechanical strength and nanoscale morphology of CNTs make them attractive for biomedical applications, particularly for developing nanofibrous bioactive surfaces in combination with HA or bioactive glasses (Chen *et al.* 2006b; Boccaccini *et al.* 2007; White *et al.* 2007; Singh *et al.* 2008). CNT layers have been shown to provide a suitable surface nanotopography for the adhesion of cells and their growth (MacDonald *et al.* 2005; Harrison & Attala 2007). Moreover, CNTs are being combined with biopolymers for the manufacture of tissue engineering scaffolds (Misra *et al.* 2007; Mwenifumbo *et al.* 2007) and to promote the formation of bone-like calcium phosphate crystals (biomineralization) when CNT coatings are in contact with relevant biological fluids (Akasaka *et al.* 2006; Aryal *et al.* 2006). Additionally,

CNTs can be used as reinforcing elements in biopolymers and bioactive ceramic matrices (Chen *et al.* 2006b; Boccaccini *et al.* 2007; White *et al.* 2007; Singh *et al.* 2008). It is also being proposed that CNTs can add functionalities to biomedical coatings, for example for improved tracking of cells, sensing of microenvironments and delivering of transfection agents besides providing nanostructured surfaces for integration with host tissue (Boccaccini *et al.* 2007). The EPD of CNTs and combinations of CNTs with HA and bioactive glass particles are presented in the next sections. EPD methods developed recently for the synthesis of functionalized single-walled CNTs/biopolymer nanocomposite films were discussed in §4.2.

5.2. EPD of CNTs

It is now well known that EPD is a very convenient technique to manipulate CNTs in suspension in order to form ordered, oriented CNT arrays, as reviewed elsewhere (Boccaccini *et al.* 2006b). The preparation of stable liquid suspensions of CNTs, in which CNTs have a high ζ -potential and the suspension has low ionic conductivity, is a requisite for successful EPD of CNTs. The stability of CNT suspensions, determined by ζ -potential measurements, has been studied mainly in aqueous and ethanol-based systems (Boccaccini *et al.* 2006b). Of significance for biomedical applications and biosensors, the high aspect ratio and surface charge of acid-treated CNTs makes them suitable scaffolds or templates for deposition of other nanoparticles, such as metallic and oxide nanoparticles, via adsorption or nucleation at the acidic sites. Moreover, the fabrication of complex patterns of CNT deposits can be realized by EPD using masks or by designing combinations of conductive and non-conductive surfaces. Composites consisting of ceramic nanoparticles and CNTs have been produced recently by sequential EPD and by electrophoretic codeposition (Boccaccini *et al.* 2010; Mahajan *et al.* 2008).

5.3. HA/CNT composites

The combination of HA with CNTs is being investigated to exploit the superior mechanical properties of CNTs to reinforce HA layers obtained by EPD (Chen *et al.* 2006b; Singh *et al.* 2006, 2008; Boccaccini *et al.* 2007; White *et al.* 2007; Lin *et al.* 2008a). An efficient mechanical reinforcement of the HA layers can be obtained if the surface of CNTs has been functionalized not only to achieve a good dispersion of CNTs in the ceramic matrix, but also to induce an ideal interface between CNTs and HA. In one of the first published reports on this subject, CNT-reinforced HA layers on Ti–6Al–4V alloy substrates were obtained using sol–gel synthesized nanosized (20–30 nm) HA particles mixed with multi-walled CNTs (MWCNTs) (Singh *et al.* 2006). Subsequently, homogeneous HA/CNT composite coating layers were obtained using different EPD processing parameters (applied voltage and deposition time) (Kaya 2008; Kaya *et al.* 2008). Lin *et al.* (2008a) deposited HA nanoparticles and

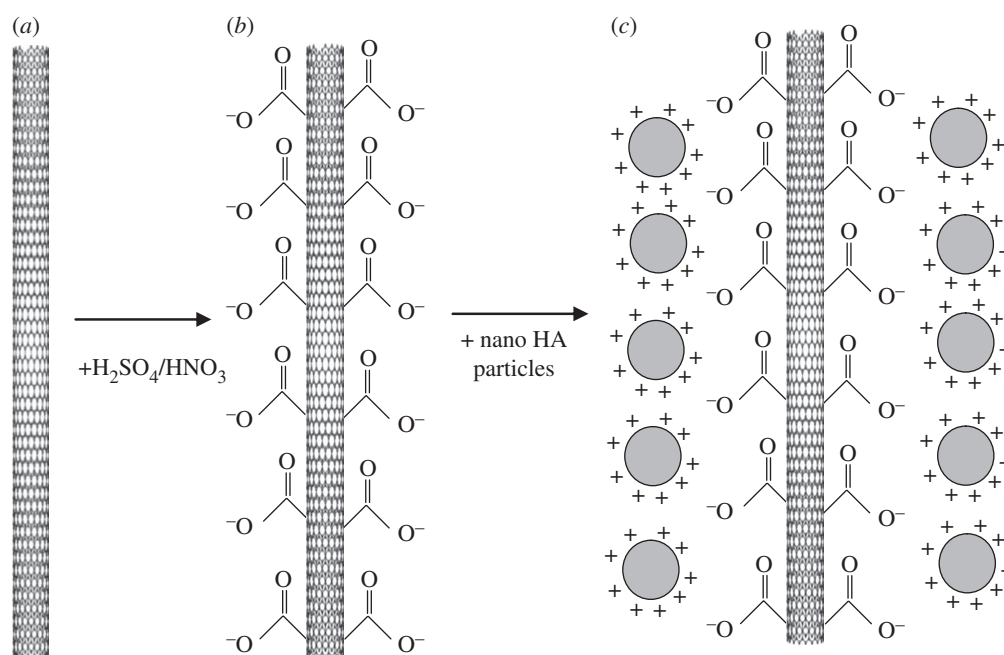


Figure 10. Schematic of the process for anchoring HA nanoparticles (positively charged) onto functionalized CNTs (negatively charged): (a) as received CNT, (b) functionalized CNT and (c) attachment of nano-HA particles onto the modified CNT surface (Boccaccini *et al.* 2010). (Reproduced with permission of Elsevier.)

MWCNTs dispersed in ethanol (with a concentration of 0.005 g ml^{-1} and a pH value of 5) on Ti alloys using an applied voltage of 30 V and a deposition time of 50 s. The resultant layer was sintered at 700°C for 2 h under flowing argon. A bonding strength of 35.4 MPa was measured between the substrate and the HA/CNT composite layer, which was higher than that achieved by pure HA layer (20.6 MPa) (Lin *et al.* 2008a). HA/CNT composites comprising HA nanoparticles (20–30 nm) and high aspect ratio MWCNTs (10–30 nm in diameter, up to $500 \mu\text{m}$ length and $40\text{--}300 \text{ m}^2 \text{ g}^{-1}$ surface area) have also been developed by EPD (Kaya *et al.* 2009). The CNTs were first treated to attach carboxylic acid groups to their surfaces. Carboxylated CNTs were then diluted in distilled water and filtered. It was anticipated that carboxylic groups induce chemical reactions along the CNT/HA interface enhancing the reinforcing efficiency of CNTs. Figure 10 represents the chemical functionalization of MWCNTs using a mixture of H_2SO_4 and HNO_3 . The negative surface charge of CNTs, owing to the presence of COOH groups, attracts positively charged HA particles (Boccaccini *et al.* 2010). In the reported study (Kaya *et al.* 2009), EPD was carried out for 4 min on Ti6Al4V substrates under constant applied voltage of 20 V DC leading to homogeneous CNT/HA composite layers. A suitable approach to yield homogeneous dispersions of CNTs in the HA matrix is to adjust the surface charge of HA nanoparticles and CNTs to be of opposite sign, so that during EPD they can attract each other and act as a single ‘composite particle’. As mentioned above, one of the main objectives for adding CNTs to HA is to increase the mechanical performance of the HA layer by inducing toughening mechanisms. In this context, the propagation of cracks induced by indentation technique on monolithic HA and on HA/CNT composites (2 wt% CNTs) has been investigated (Kaya 2008; Kaya *et al.* 2008, 2009), and it

was shown that no crack deflection occurs but possible toughening mechanisms such as nanotube crack bridging and debonding appear to be active.

Another convenient alternative to functionalize CNTs for EPD is based on hydrothermal processing. This technique has the advantage over the conventional acid treatment method described above (using a HNO_3 and H_2SO_4 mixture) that it will not introduce defects on the nanotubes that could have a detrimental effect on the CNTs’ mechanical and functional properties. Under hydrothermal conditions surface groups such as ($-\text{COOH}$) and ($-\text{OH}$) attach on to the surfaces of CNTs, resulting in good CNT dispersion and formation of stable suspensions (Kaya *et al.* 2010). As in all CNT-containing materials, issues related to the possible cytotoxicity of CNTs should be investigated thoroughly and clarified in future investigations. There is increasing evidence that CNTs under certain conditions may be toxic to cells (Jia *et al.* 2005; Harrison & Attala 2007; Firme & Bandaru 2010; Zhang *et al.* 2010); however, more comprehensive work is still required to confirm these effects and to ascertain the conditions at which CNT may be toxic (both *in vitro* and *in vivo*). In this regard, CNT coatings obtained by EPD may represent reliable and reproducible CNT substrates to be used as model systems to investigate CNT cytotoxicity.

5.4. CNT/SiO₂ coatings

The combination of silica nanoparticles with MWCNTs by EPD for potential application in the biomedical field was investigated by Chicatun *et al.* (2007), who prepared SiO₂/CNT nanoporous composite films on metallic substrates. Commercially available MWCNTs were used without any post-synthesis treatment and distilled water was the solvent. An aqueous dispersion of hydrophilic fumed silica was considered as the

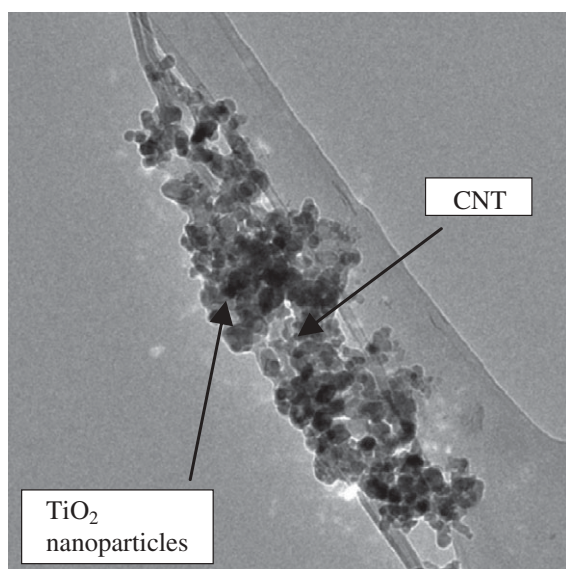


Figure 11. TEM micrograph showing the interaction between TiO_2 nanoparticles and CNTs in aqueous suspensions at $\text{pH} < 5$ (Singh *et al.* 2006).

source of SiO_2 . To prepare optimal CNT suspensions in distilled water, the surface properties of CNTs were modified with Triton X-100 and iodine. The suspension for electrophoretic codeposition of CNTs and SiO_2 nanoparticles was prepared by mixing an aqueous CNT suspension with different volume ratios of the fumed silica dispersion. Layered CNT/ SiO_2 porous composites were obtained by sequential EPD experiments, alternating the deposition of CNTs and SiO_2 nanoparticles. Related developments were carried out by Wang & Huang (2008), who demonstrated the fabrication of novel Ag/ SiO_2 /CNT composites by pulsed voltage electrophoretic codeposition.

5.5. CNT/ TiO_2 coatings

The fabrication of CNT/ TiO_2 composites by EPD has been explored considering the expected benefit of combining these two materials in terms of functional and mechanical properties (Cho *et al.* 2008; Jarernboon *et al.* 2009). Homogeneous and thick deposits of CNTs have been coated and infiltrated with TiO_2 nanoparticles by EPD using commercially available TiO_2 nanoparticles (P25, Degussa, Frankfurt, Germany) (Cho *et al.* 2008). The first research work in this particular system was carried out using MWCNTs, grown by chemical vapour deposition, and commercial TiO_2 nanopowder (Singh *et al.* 2006). For codeposition of CNTs and TiO_2 nanoparticles on stainless steel substrate, suspensions were prepared by mixing 3.5 g of TiO_2 nanoparticles with 31.5 g of CNT aqueous solution. It was shown that codeposition has occurred owing to the opposite signs of the surface charges of CNTs and TiO_2 nanoparticles at the working pH. As discussed above for the case of CNT/HA composites, the electrostatic attraction between CNTs and TiO_2 particles leads to the formation of composite particles in suspension consisting of TiO_2 nanoparticles covering the surface of individual CNTs, as seen in figure 11

(Singh *et al.* 2006). The process of EPD at constant voltage conditions was optimized to achieve homogeneous and well-adhered deposits of varying thicknesses on the metallic substrates. In a following investigation by Cho *et al.* (2008), both the sequential deposition of alternating layers of CNTs and TiO_2 nanoparticles and the codeposition of TiO_2 /CNT composites were investigated. The different strategies suggested for the EPD of TiO_2 /CNT nanocomposite coatings are schematically shown in figure 12 (Boccaccini *et al.* 2010). In the particular case of sequential deposition, once a porous CNT coating has been obtained, EPD is applied to incorporate titania nanoparticles or to fabricate layered structures (figure 12a; Boccaccini *et al.* 2010). Alternatively, composite CNT/ TiO_2 nanoparticle coatings can be obtained by electrophoretic codeposition from stable suspensions containing CNTs and TiO_2 nanoparticles. The various components may be separately dispersed or may be preassembled to form more complex building blocks in suspension (figure 12b). It should be mentioned that if larger particles are used, CNTs can be made to wrap individual particles forming charged CNT-coated particulates in suspension.

The mechanisms involved in electrophoretic codeposition of CNTs and nanoparticles of the same charge sign have been explained by considering the different trajectories of nanoparticles in suspension (Cho *et al.* 2008). The presence of CNTs has been confirmed to reduce the problem of microcracking in TiO_2 films by bonding to TiO_2 nanoparticles through the interaction of hydroxyl and carboxyl groups (Jarernboon *et al.* 2009), which can effectively lead to strong CNT/ TiO_2 coatings for biomedical applications. Despite the promise of these nanostructured coatings for biomedical applications, to the authors' knowledge, studies of the biocompatibility of such coatings have not been carried out as yet.

5.6. Bioactive glass/CNT composites

EPD has been considered an alternative technique to plasma-spraying or enamelling to deposit bioactive glass coatings on metallic substrates (as discussed in §2.2) and, more recently, the EPD technique was further adapted for the fabrication of CNTs/Bioglass-layered composite films (Meng *et al.* 2009; Schausten *et al.* 2010). The bioactive glass powder used has a particle size between 1 and 15 μm . After acid treatment of CNTs, a well-dispersed aqueous CNT suspension with a concentration of 0.45 mg ml^{-1} was prepared. Sequential EPD was applied to produce a coating of CNTs on bioactive glass layers (Cho *et al.* 2009) using electric field strengths in the ranges 10–40 and 2–10 V cm^{-1} for CNTs and Bioglass suspensions, respectively, and different deposition times (between 1 and 6 min). A similar technique has been used to deposit CNTs onto three-dimensional Bioglass and polyurethane scaffolds by electrophoretic impregnation (Boccaccini *et al.* 2007; Meng *et al.* 2009; Zawadzak *et al.* 2009). SEM images showing the typical microstructure of a CNT-coated Bioglass-based scaffold, obtained by EPD (2.8 V, 10 min), are shown in figure 13 (Meng *et al.* 2009). Several applications of these novel

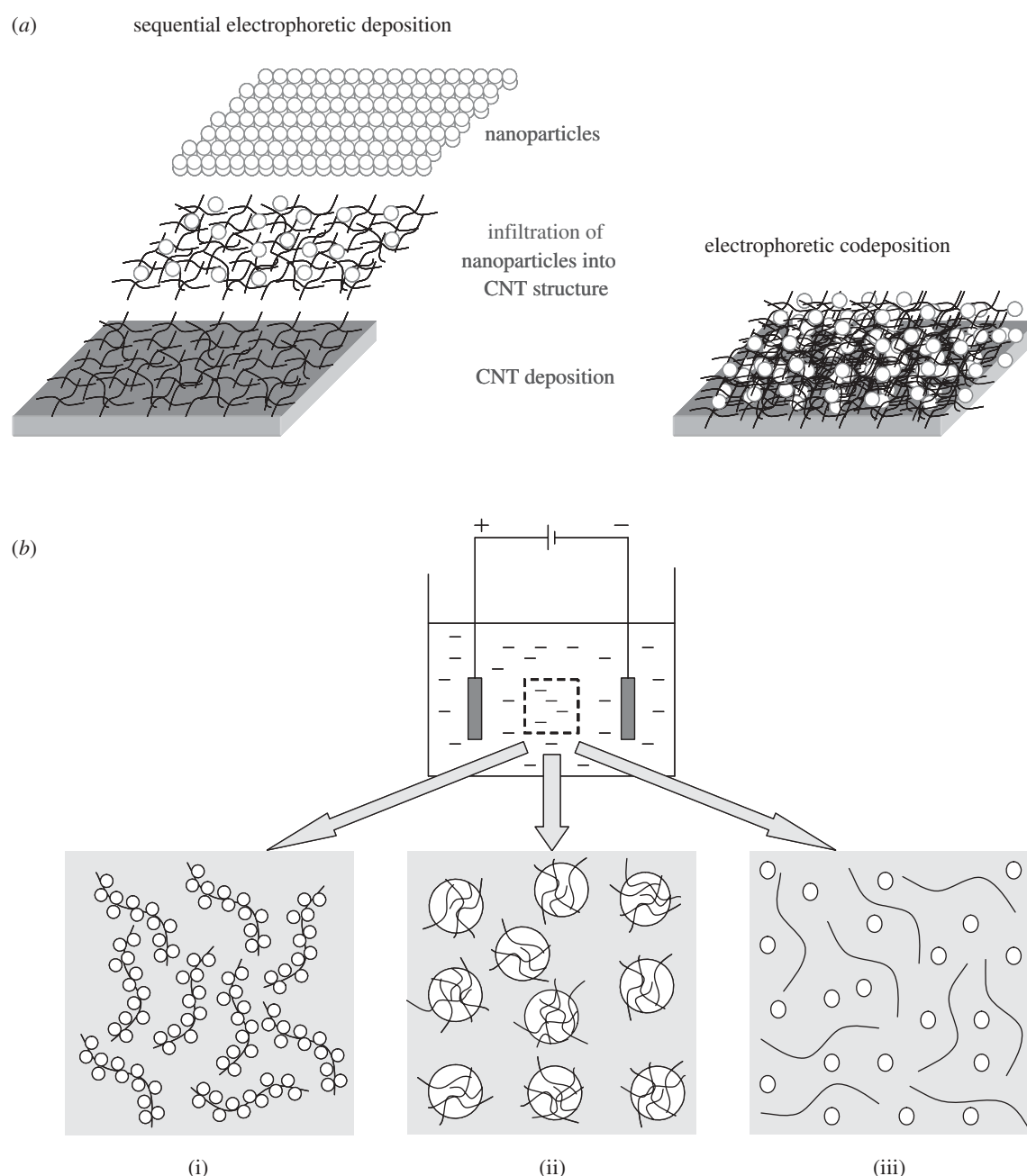


Figure 12. Different approaches to fabricate CNT/ceramic nanocomposites by EPD (Boccaccini *et al.* 2010): (a) schematic diagram showing sequential deposition of CNTs and nanoparticles form layered heterostructures and electrophoretic codeposition; (b) schematic diagram showing different alternatives to produce CNT composites by electrophoretic codeposition: (i) self-assembly of nanoparticles on individual CNTs, (ii) heterocoagulation of CNTs onto individual (larger) particles, and (iii) simultaneous deposition of CNTs and ceramic particles with the same charge polarity in suspension. (Reproduced with permission of Elsevier.) Plain thin curves, CNTs; open circles, ceramic or metallic particles.

nanocomposites are possible, especially in orthopaedics and bone engineering. The nanoporous network provided by the CNT deposited on the Bioglass surface is a suitable substrate to study cell attachment and growth. The presence of CNTs should enhance the bioactive function for applications as coatings on orthopaedic implants for strong bonding with bone, similarly as discussed above for HA/CNT composites. In the field of scaffolds for bone tissue engineering, CNT-coated Bioglass substrates produced by EPD should enable the rapid growth of nano-HA crystals when immersed in SBF (Akasaka *et al.* 2006; Aryal *et al.* 2006), thus providing a favourable surface for

osteoblast cell attachment and proliferation. As mentioned above, issues related to the biocompatibility of CNT/Bioglass composites in contact with host cells remain to be investigated, including possible cytotoxicity effects of CNTs (Jia *et al.* 2005; Firme & Bandaru 2010; Zhang *et al.* 2010).

6. OTHER APPLICATIONS OF EPD IN NANOBIO TECHNOLOGY

EPD is considered a suitable method for nanofabrication, including the development of nanostructures

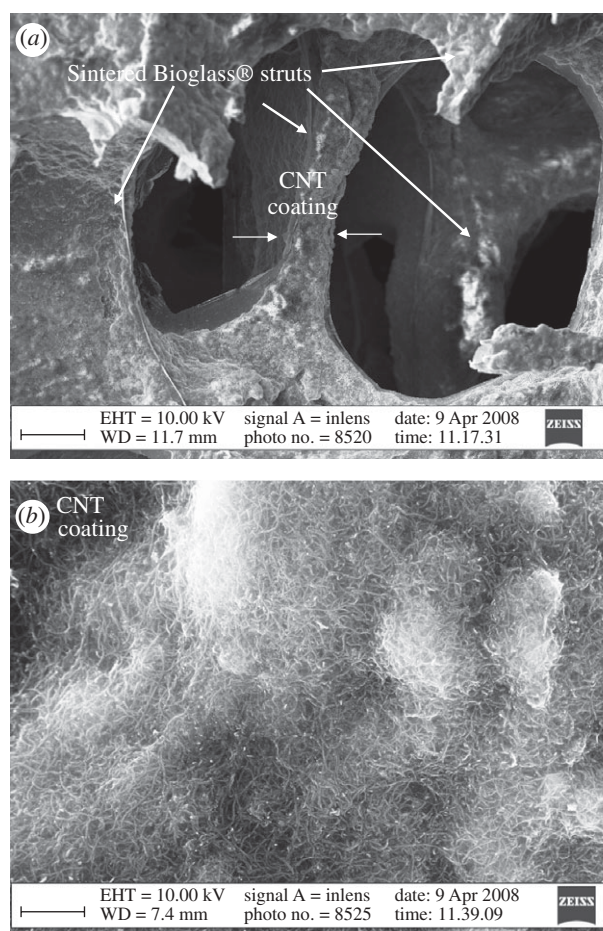


Figure 13. SEM images showing the typical microstructure of a CNT-coated Bioglass-based scaffold, obtained by EPD (2.8 V, 10 min) at different magnifications (Meng *et al.* 2009). Scale bars, (a) 100 μm ; (b) 1 μm . (Images courtesy of Dr Decheng Meng.)

and devices, such as nanomachines and components for nanoelectronics with potential applications in nanomedicine and drug delivery systems. A few examples are described in this section to illustrate the potential of EPD in this field. For example, Iwata *et al.* (2007) have investigated EPD to deposit gold nanoparticles (dots) on Si surfaces from a nanopipette probe filled with the deposition suspension. The nanopipette probe was the EPD cell and the two electrodes were made of a thin metal wire placed inside the nanopipette and a conductive surface in contact with the edge of the nanopipette. When a difference of potential was applied between the electrodes, the colloidal particles migrated towards the edge of the probe and deposited on the surface. The size of the Au dots could be tailored by changing the deposition time and the voltage during EPD. An advanced application of EPD to fabricate a flexible drug delivery device was shown by Huang *et al.* (2009). Drug-carrying magnetic core-shell $\text{Fe}_3\text{O}_4/\text{SiO}_2$ nanoparticles were electrophoretically deposited onto an electrically conductive flexible polyethylene terephthalate (PET) substrate. The PET substrate was first patterned to a desired layout and subjected to deposition. A uniform nanoporous membrane could be produced. After lamination of the

patterned membranes, a final chip-like device was formed that can be used for controlled delivery of drugs. The release of useful drugs can be controlled by directly modulating the magnetic field. The flexible and membrane-like drug delivery chip uses drug-carrying magnetic nanoparticles as the building blocks that ensure a rapid and precise response to magnetic stimulus (Huang *et al.* 2009). Another advanced biomedical device based on an EPD-fabricated structure was presented by So *et al.* (2007), who developed a drug-eluting stent (DES) with antiproliferative agents. They electrophoretically coated poly(lactic-co-glycolic acid) (PLGA) nanoparticles embedded with curcumin onto a metal stent for local drug delivery. Polyacrylic acid was used as surfactant because its carboxylic group contributes negative charges to the surface of the PLGA nanoparticles. The study of So *et al.* (2007) was thus the first to demonstrate that electrophoretic coatings with functional polymeric nanoparticles represent an attractive technology with a wide application in DES as various kinds of nanoparticles and drugs can be used, exploiting the versatility of EPD.

The well-known EPD of charged polymer colloids (Boehmer 1996; Hayward *et al.* 2000), adapted to obtain colloidal monolayers on patterned electrodes (Dziomkina *et al.* 2005), could represent another opportunity for the application of EPD in the biomedical and pharmaceutical fields. For example, it was suggested that the technique can be used for the preparation of three-dimensional colloidal crystals by means of layer-by-layer deposition of oppositely charged colloidal particles or to grow colloidal crystals with different lattice structures, which is determined by the patterns of the electrode substrates. Applications of patterned colloidal monolayers to simulate the behaviour of hollow microspheres (capsules) used as drug delivery systems or for protection of sensitive substances such as enzymes and proteins are suggested (Dziomkina *et al.* 2005).

EPD of biological entities, including enzymes, bioactive molecules, bacteria and cells, is receiving increasing research interest in the fields of biotechnology and biomedicine. The growth and positioning of collagen membranes, for example, was investigated by Baker *et al.* (2008). Collagen, when net charged, migrates in an electric field like any charged polymer. However, collagen behaviour in an electric field has not been widely investigated, thus representing an interesting area for future investigation in the field of biopolymer coatings. It has been shown (Baker *et al.* 2008) that the electric field and the pH can be used in conjunction to produce suspended collagen membranes from solutions of type I collagen monomers. Such membranes can be deposited on an electrode or, as shown by Baker *et al.* (2008), the spatial aggregation of collagen in an electrolyte is possible in order to develop highly reproducible self-supporting collagen films.

Immobilization of enzymes, which is of high interest for the development of biosensors, can be achieved by electrophoresis, which should enable the deposition of enzymes in a highly active state. Recent developments have shown the application of unbalanced AC fields of sufficient amplitude to achieve deposition of enzymes

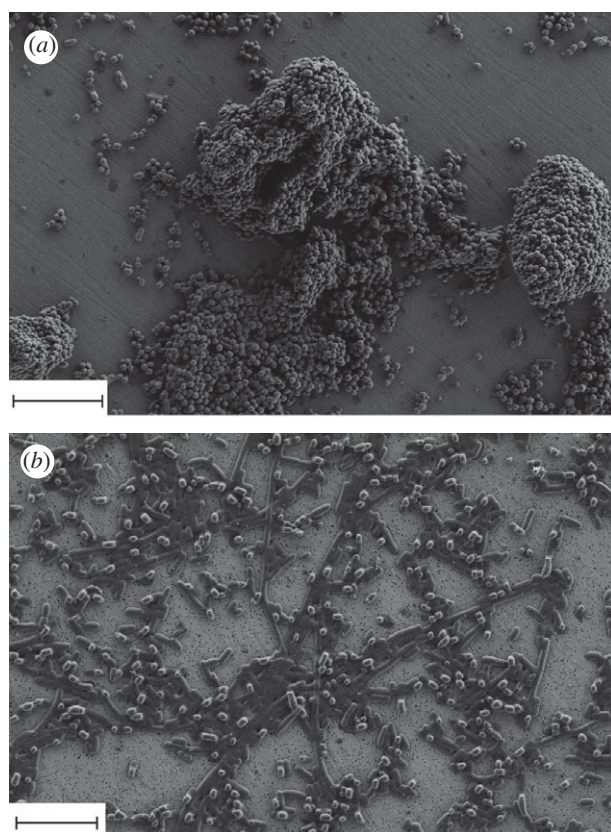


Figure 14. SEM images showing (a) inclusion bodies electrophoretically deposited on steel cathode from suspensions at IEP of *E. coli* bacteria (3 min of EPD at 7.5 V cm^{-1}) and (b) deposition of *E. coli* on the anode after the pH was set to the IEP of the inclusion bodies (Novak *et al.* 2009). Scale bars, (a,b) $10 \mu\text{m}$. (Reproduced with permission of Elsevier.)

and bacteria (Ammam & Fransaer 2009; Neirinck *et al.* 2009b). Ammam & Fransaer (2009), for example, deposited glucose oxidase from water onto a platinum electrode. Glucose oxidase layers with a thickness of $7 \mu\text{m}$ could be obtained after 20 min of EPD. In contrast, if a symmetrical alternating signal was used under the same conditions, a layer of only $0.5 \mu\text{m}$ was formed. Two effects involved in the deposition were indicated: the electrophoretic migration of the enzyme towards the deposition electrode and the pH-induced precipitation of the enzyme near the deposition electrode. The effect of amplitude, frequency, deposition time and concentration on deposition rate was studied. The formation of thick enzyme layers is highly interesting for applications in biosensor technology and in other biotechnology areas (Ramanathan *et al.* 1997). In these applications, the manipulation of biological materials is challenging owing to the size and sensitivity to environmental parameters (Neirinck *et al.* 2009b). In recent experiments, Gram-negative *Escherichia coli* and Gram-positive *Staphylococcus aureus* strains were selected as model bacteria for deposition on stainless steel foil by asymmetric alternating field-based AC EPD (Neirinck *et al.* 2009b). A significant number of bacteria were found to remain viable after deposition. The work has followed from previous research in which the design of biofilms from controlled AC EPD of bacteria was demonstrated (Poortinga *et al.* 2000;

Brisson & Tilton 2002). By applying a sufficiently high current density, bacteria clusters could be immobilized in an ordered fashion on the surface of interest. Bacteria biofilms can also be used in biomedical applications, e.g. probiotic bacterial could be made to attach on biomaterials to serve as protective coatings on catheters or other devices (Reid *et al.* 1997; Poortinga *et al.* 2000) in order to inhibit the adhesion of pathogenic micro-organisms. It was also shown (Brisson & Tilton 2002) that bacteria arrays (e.g. *Saccharomyces cerevisiae*) may be induced to form geometric patterns by focusing the electric field during deposition. An interesting AC field alignment effect has been observed when rod-like bacteria were deposited (Poortinga *et al.* 2000). Moreover, Novak *et al.* (2009) have shown that EPD is a suitable tool for separation of protein inclusion bodies from host bacteria (e.g. *E. coli*) in suspension. It was shown that the efficiency of separation and the yield depend not only on the electrokinetic mobility of the species but also on electrode composition and surface morphology. This method requires precise knowledge and control of the zeta potential of the species involved. Figure 14 shows SEM images of inclusion bodies electrophoretically deposited on steel cathode from suspensions at IEP of *E. coli* bacteria (3 min of EPD at 7.5 V cm^{-1}) and the deposition of *E. coli* on the anode after the pH was set to the IEP of the inclusion bodies (Novak *et al.* 2009).

Investigations about the electrophoretic nanoprinting of proteins on conducting and non-conducting substrates, recently carried out by combining capillary electrophoresis, which enables control of protein movement, and atomic-force microscopy (Lovsky *et al.* 2010) are potentially very attractive for biotechnology (protein chips), molecular electronics and for fundamental investigations in cell biology. The technique, atomic-force-controlled capillary electrophoretic printing, has been described by Lovsky *et al.* (2010). A relevant review covering experimental and theoretical studies of cell electrophoresis mobility and its relevance in the fields of drug discovery and medicine has been published (Akagi & Ichiki 2008). The main interest in the field of cell electrophoretic mobility determination is to use it as a non-invasive marker of cell biological condition, since it reflects the electrical and mechanical properties of the cell surface. Considering both the similarity and the differences in the conditions at the surface of colloidal particles and cells, as shown in figure 15 (Akagi & Ichiki 2008), it is possible to ascertain that knowledge of the electrophoretic mobility of cells will be also relevant for the application of EPD to manipulate cells in an electric field. The surfaces of biological cells are different from that of solid colloidal particles in that charges are fixed in an extracellular layer of proteins protruding from the membrane bilayer boundary, i.e. the glycocalyx. In a complete model of the behaviour of cells under electric fields, the penetration of electrolyte ions into the glycocalyx must be considered, which poses a complexity and a significant difference with the electrophoresis of solid colloidal particles. In this case, EPD could be adapted to become a potentially useful bioprocessing tool for cell therapy

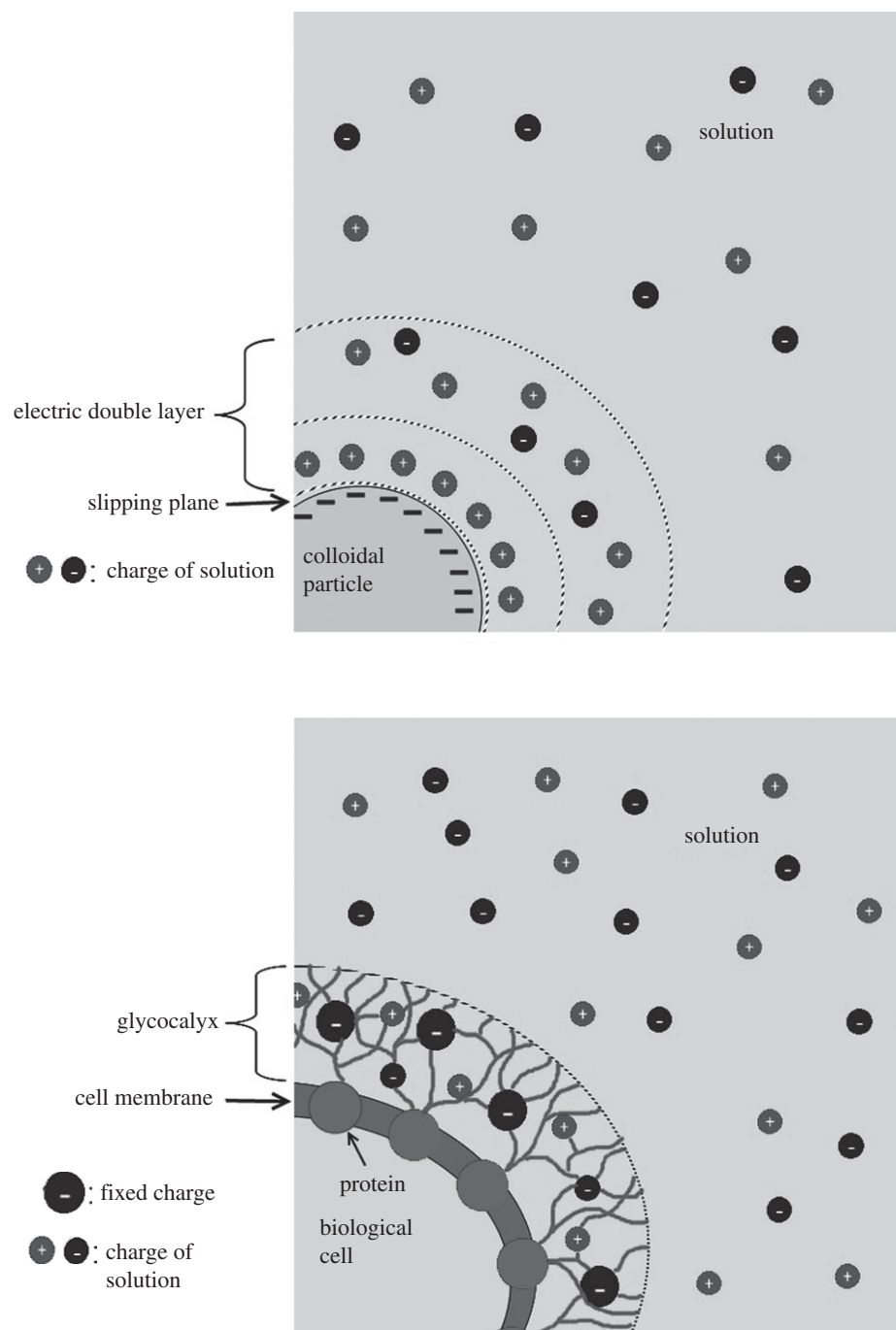


Figure 15. Schematic diagram showing different conditions at the surface of colloidal particles and cells, as suggested by Akagi & Ichiki (2008).

and tissue engineering applications. This last aspect, EPD-assisted processing of biological cells, remains an important field of research for the future.

7. CONCLUSIONS

EPD is an attractive material-processing technique characterized by its versatility in terms of the broad range of materials (in particulate form) that it can be applied to and the relatively simple equipment required. EPD offers control over nano- and microstructure, stoichiometry, microscopic and macroscopic dimensions and properties of the materials produced. Based on

these advantages, EPD is finding increasing application in the field of biomaterials. The main (and traditional) application field is in the development of bioactive coatings for orthopaedic applications, in particular HA and bioactive glass coatings. In the last few years, however, there has been a considerable increase in research efforts on the development of EPD to produce polymer–nanoceramic composite coatings with enhanced functionalities, e.g. drug delivery capability, electrical conductivity, encapsulation of proteins, antibacterial coatings and for manipulating biological entities, including enzymes and bacteria. The review of the literature has also highlighted that EPD is a suitable tool to manipulate nanomaterials in suspensions, in

particular nanoparticles and CNTs, and also for the development of biomedical nanostructures. This field is likely to be a focus of more intensive research efforts in the near future, considering the advantages of EPD, also discussed in this review. In this regard, EPD is a potentially powerful method to produce CNT-based devices, including tissue engineering scaffolds, particularly considering that few alternative techniques exist to deposit and align CNTs on three-dimensional porous structures required in tissue engineering. Of importance for the biomedical field for fabrication of biomaterial structures, EPD has the significant advantage, compared with other fabrication routes, of operating at low temperatures, and it can be easily scaled up using inexpensive equipment. EPD has thus the potential to lead to commercial success and large-scale production. Despite its advantages and the large range of successful applications of EPD in the biomaterials field, further theoretical and modelling work is required to gain complete understanding of the mechanisms involved in EPD, especially when combinations of biopolymers, nanoparticles and biomolecules are considered. It has become apparent that optimization of EPD parameters is usually carried out by time-consuming trial-and-error approaches. This unsatisfactory situation is the result of lack of available quantitative relationships linking EPD process parameters, deposition kinetics and final deposit properties in most cases. From the applications point of view, the specific areas in which EPD is expected to expand, particularly in the field of bionanotechnology, are fabrication of nanostructured and hybrid composite biomaterials, functionally graded bioactive and biomimetic coatings, CNT-containing devices as well as the development of biopolymer composites encapsulating biomolecules and drugs for multi-functional coatings, biosensors and for regenerative medicine. The adaption of EPD as a bioprocessing method for the direct manipulation of cells, which could be highly relevant in cell therapy approaches, cell-based drug screening and in the general regenerative medicine field, remains also a goal for future highly interdisciplinary investigations.

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